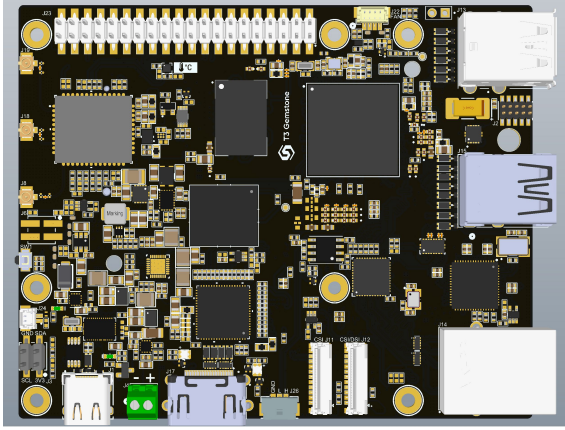


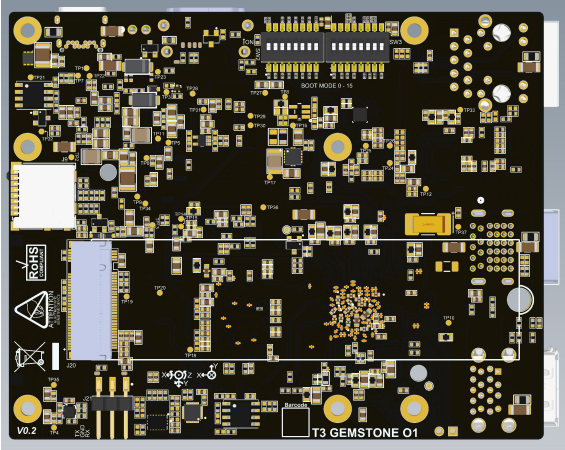
T3_Gemstone

Rev : V0.2

TOP VIEW



BOTTOM VIEW



REV

DATE

DESCRIPTION

V0.1

23.07.2024

Initial release.

V0.2

15.07.2025

**T3 Gemstone**

Proje / Project:

T3_Gemstone

Sayfa Adı / Page Name:

01-temp_title.SchDoc

Revizyon / Revision:

V0.2

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1 / 33

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T3 VAKFI

Tarih / Date

15.07.2025

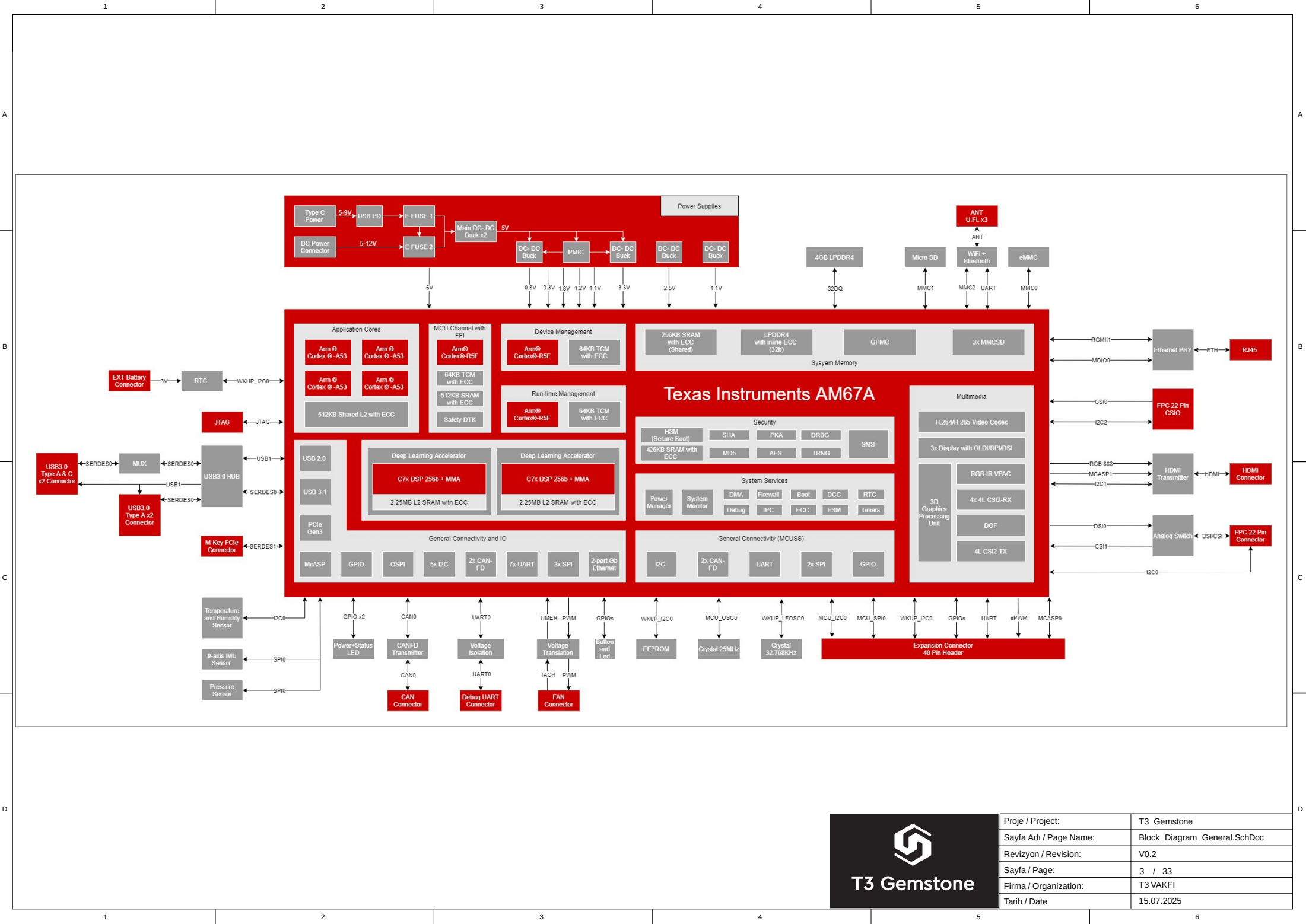
Page	Page Title
01	Title
02	Table_of_contents
03	Block_Diagram_General
04	Block_Diagram_Power
05	Block_Diagram_I2C
06	Power_Input
07	Main_Power_Rails
08	PMIC
09	Power_Distribution
10	SOC_Power
11	SOC_GPIO_Power - DDR_Power
12	SOC_Digital_Power
13	SOC_Ground - JTAG
14	SOC_MCU_Domain_GPIO -WKUP-Resets
15	SOC_MMC - SD CARD
16	SOC_DDR - OSPI
17	SOC_GPMC - VOUT
18	SOC_SERDES - USB
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A

B

C

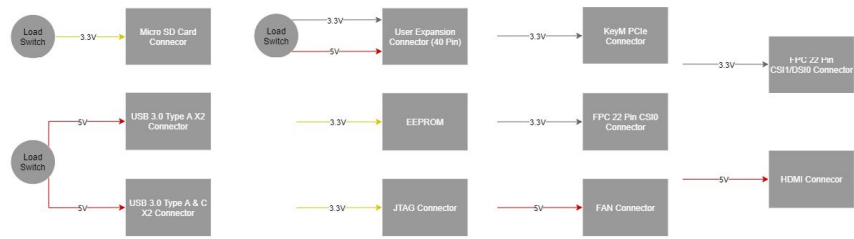
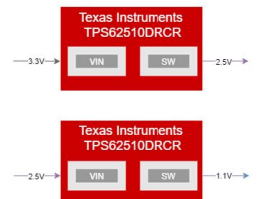
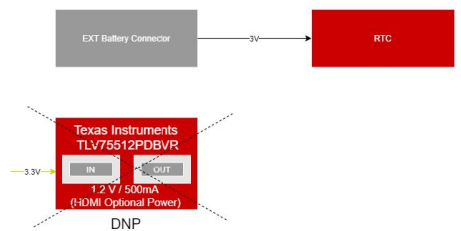
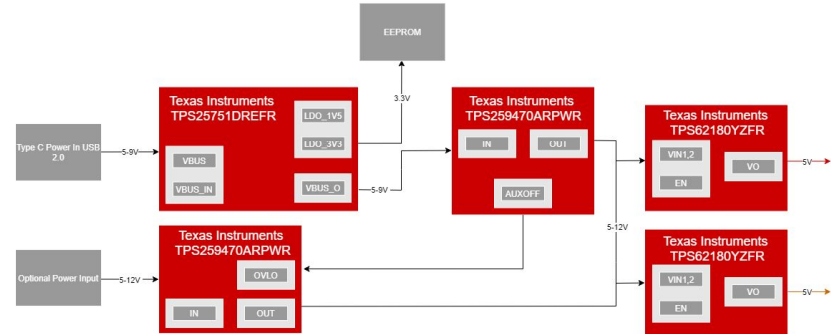
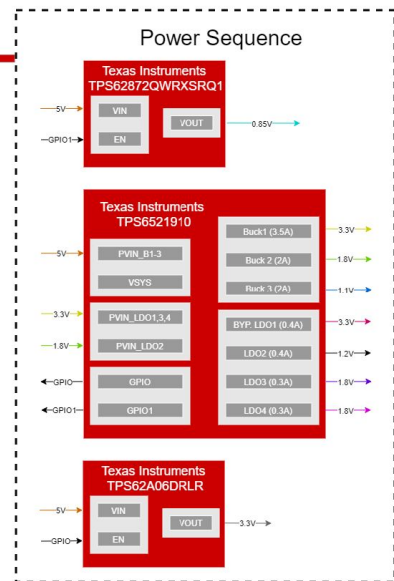
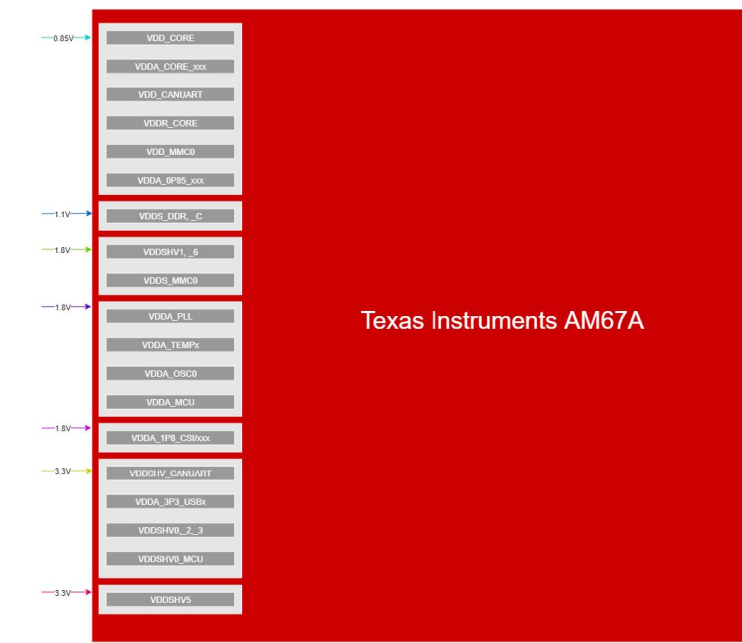
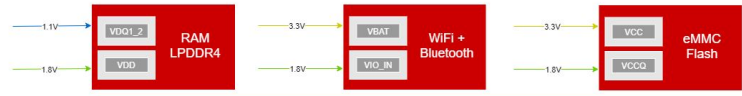
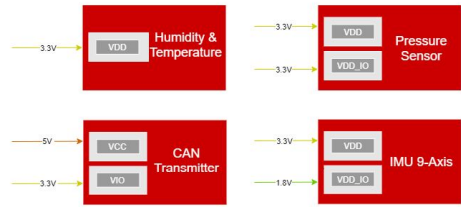
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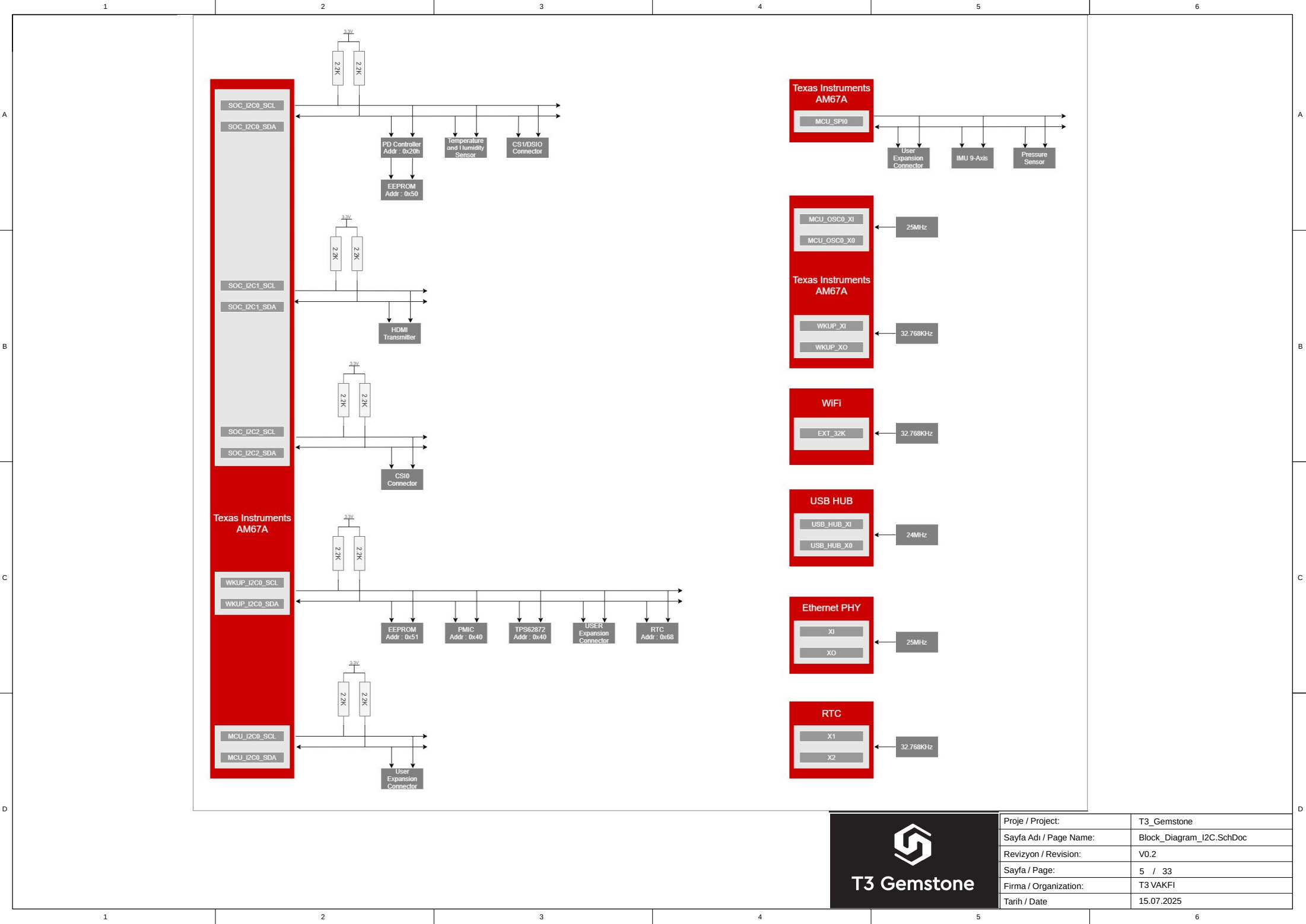
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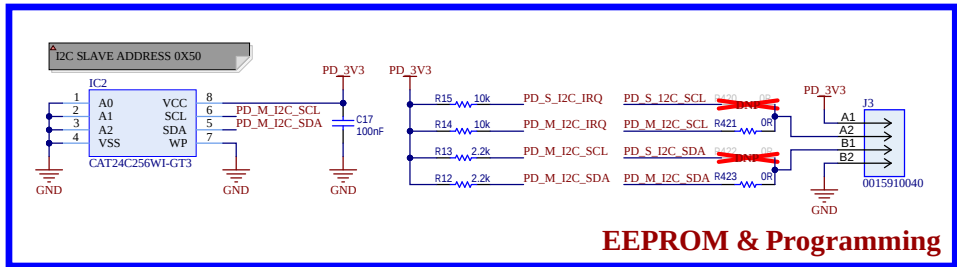
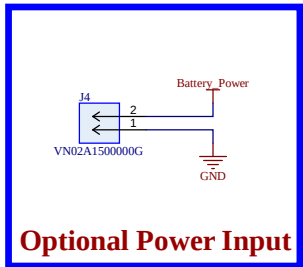
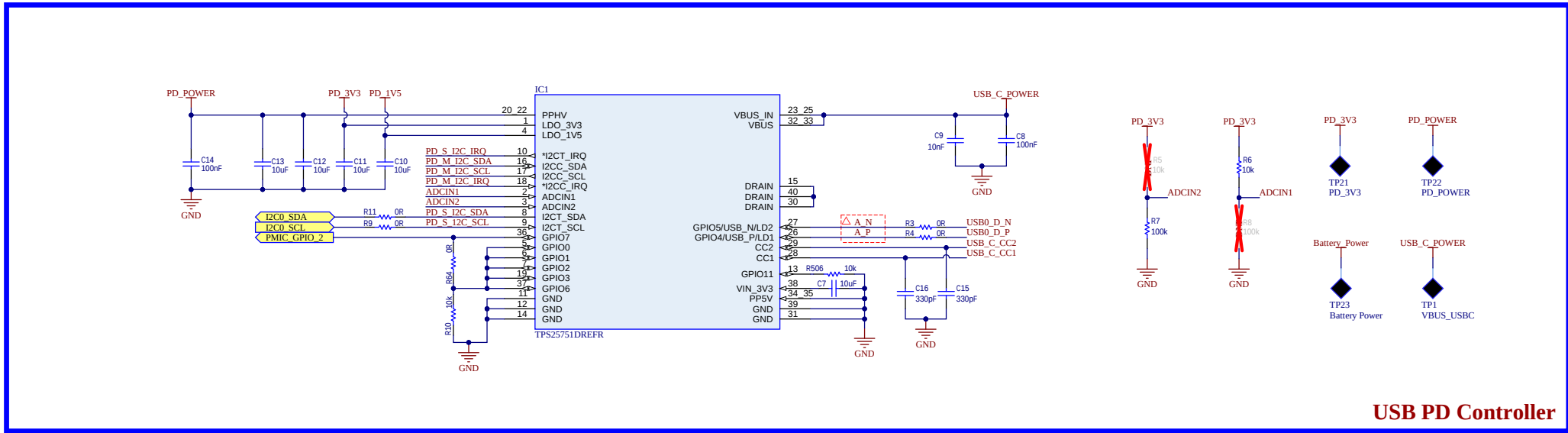
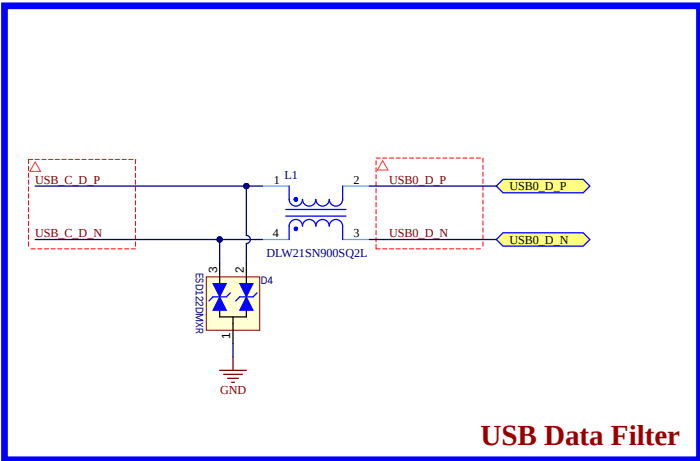
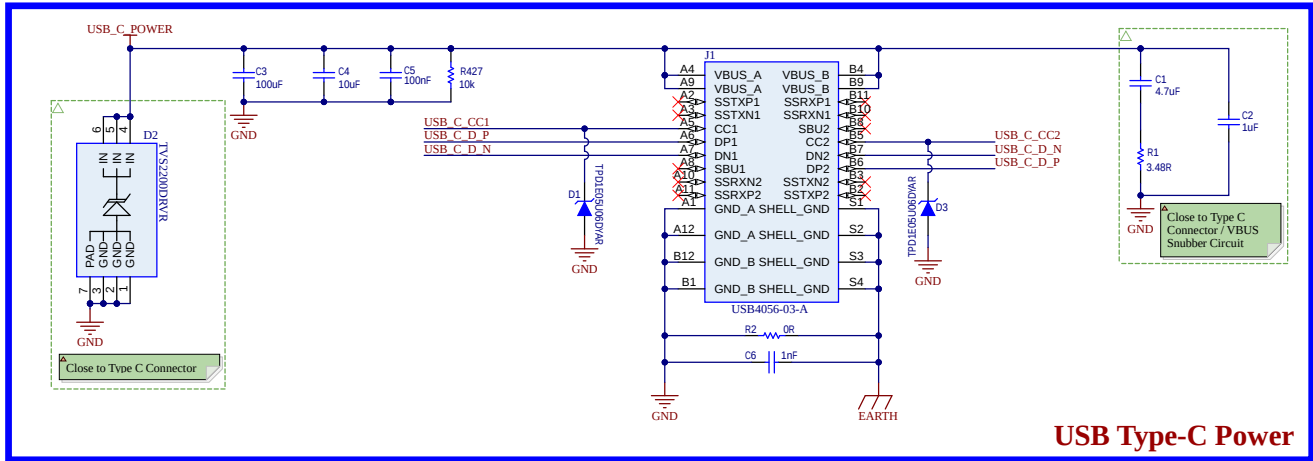
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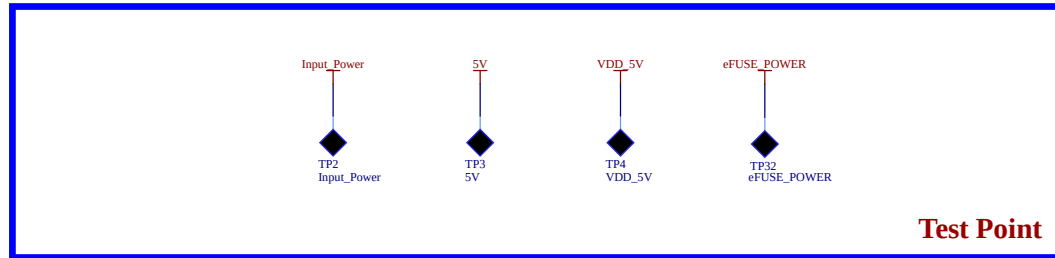
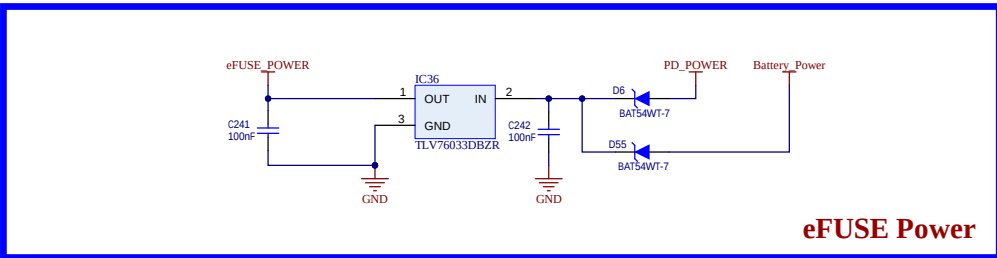
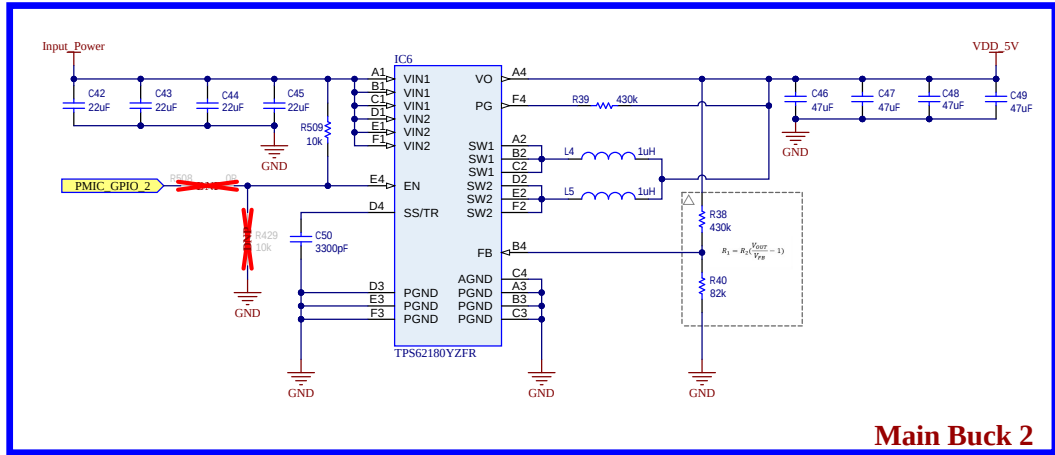
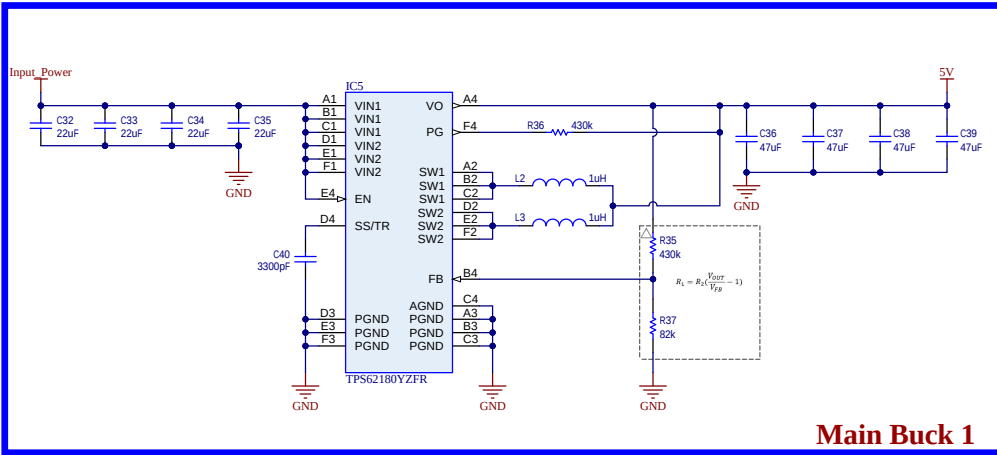
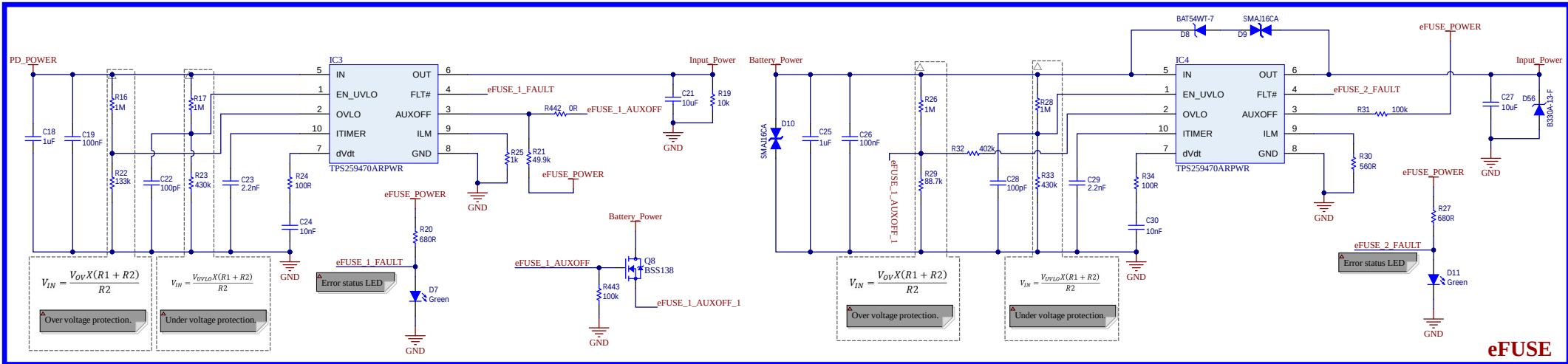


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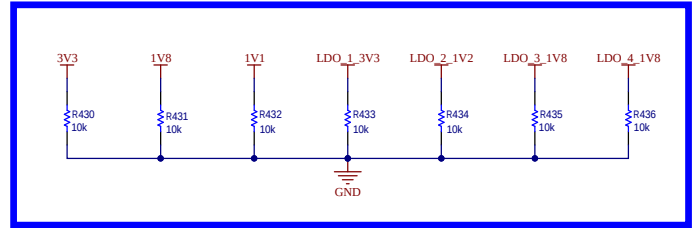
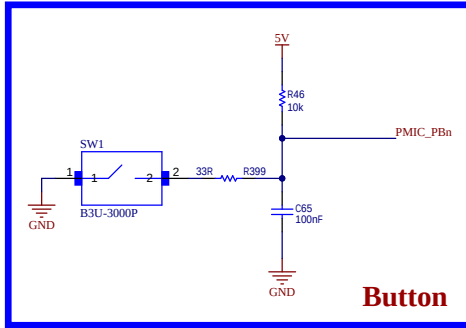
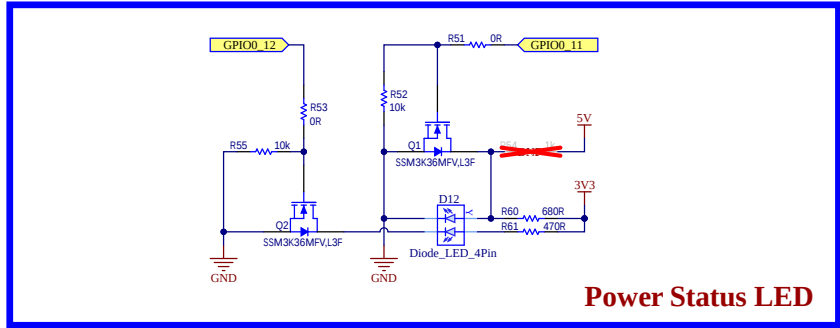
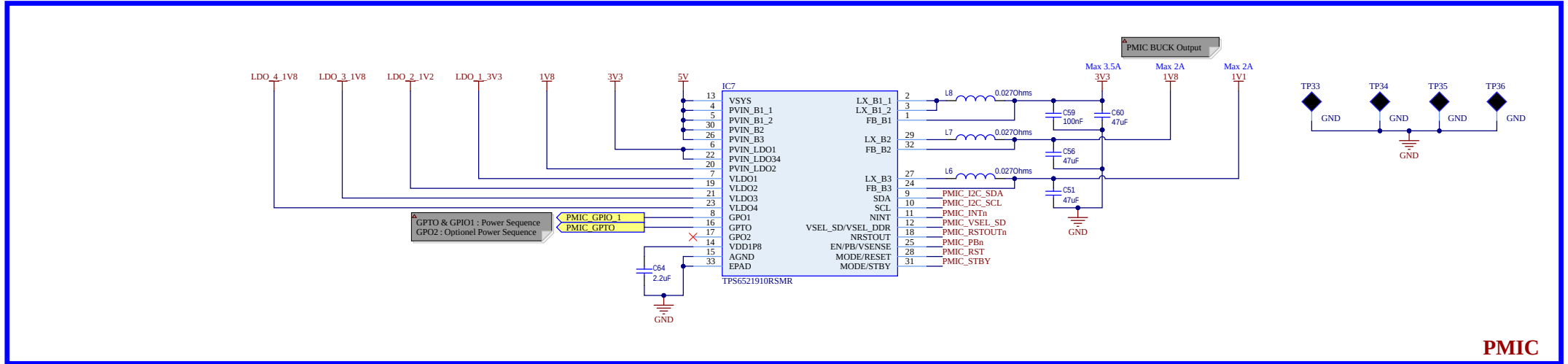
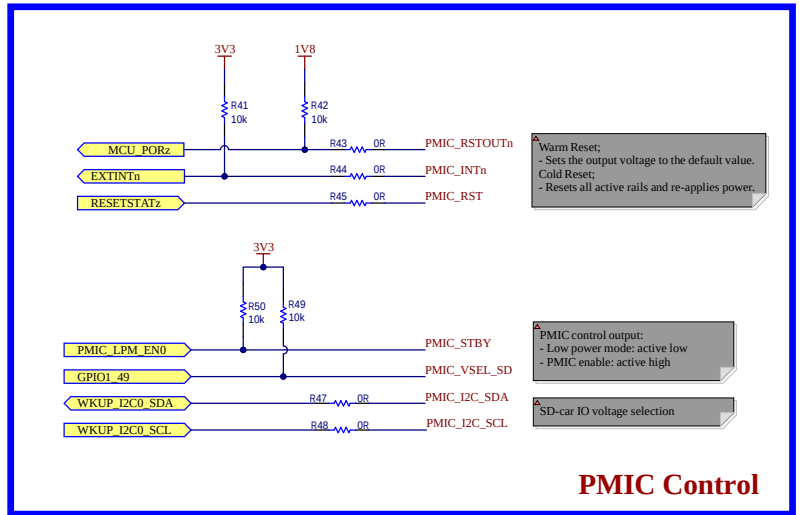
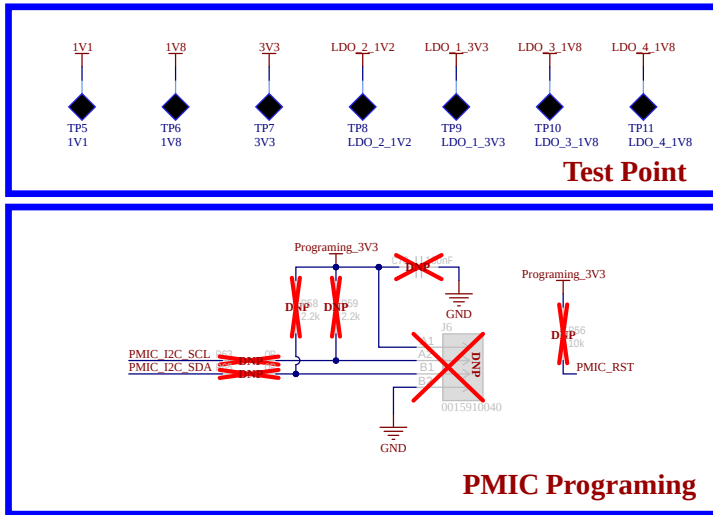
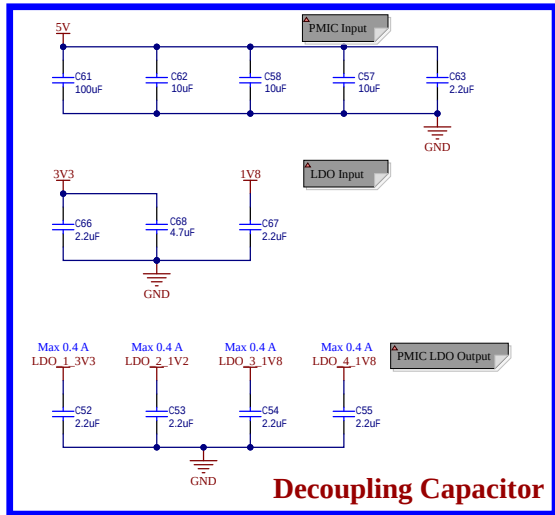


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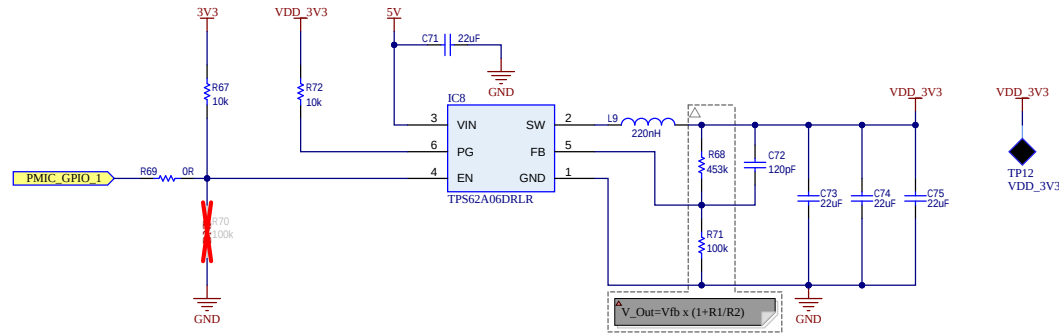


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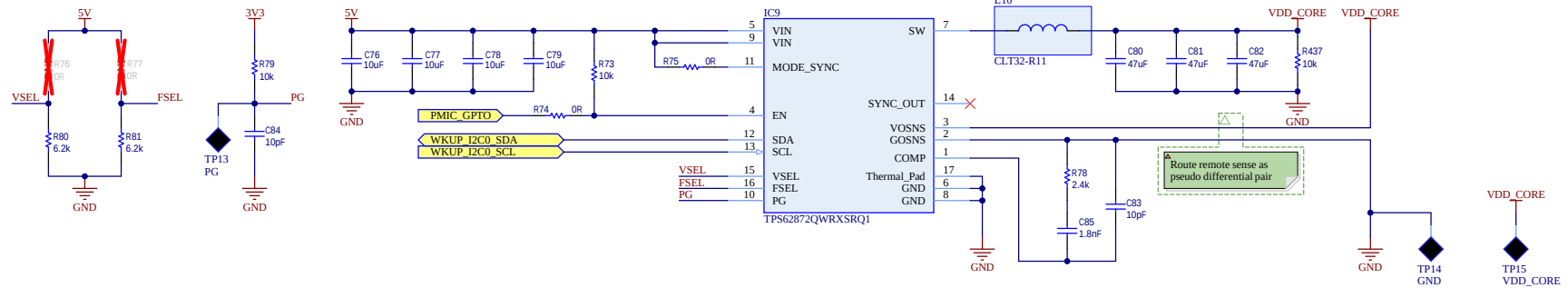
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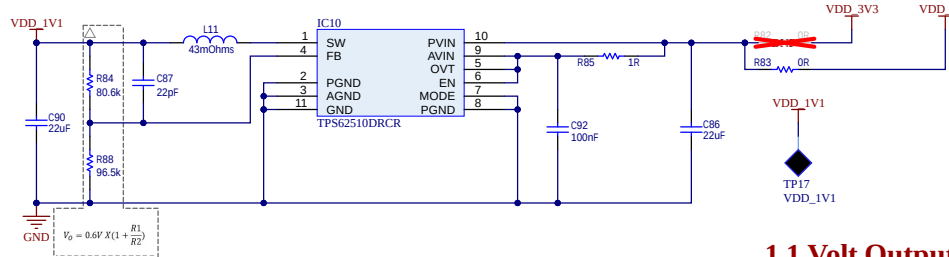
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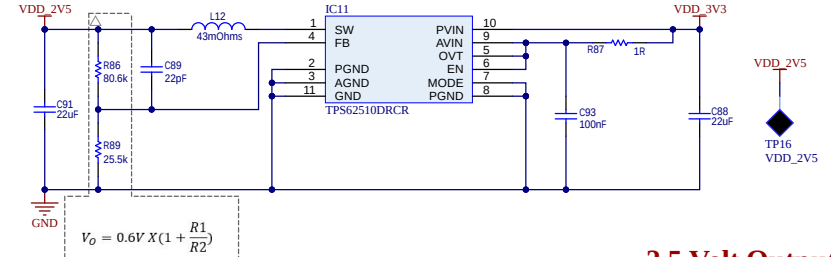
3.3 Volt Output



SoC Supply Output



1.1 Volt Output

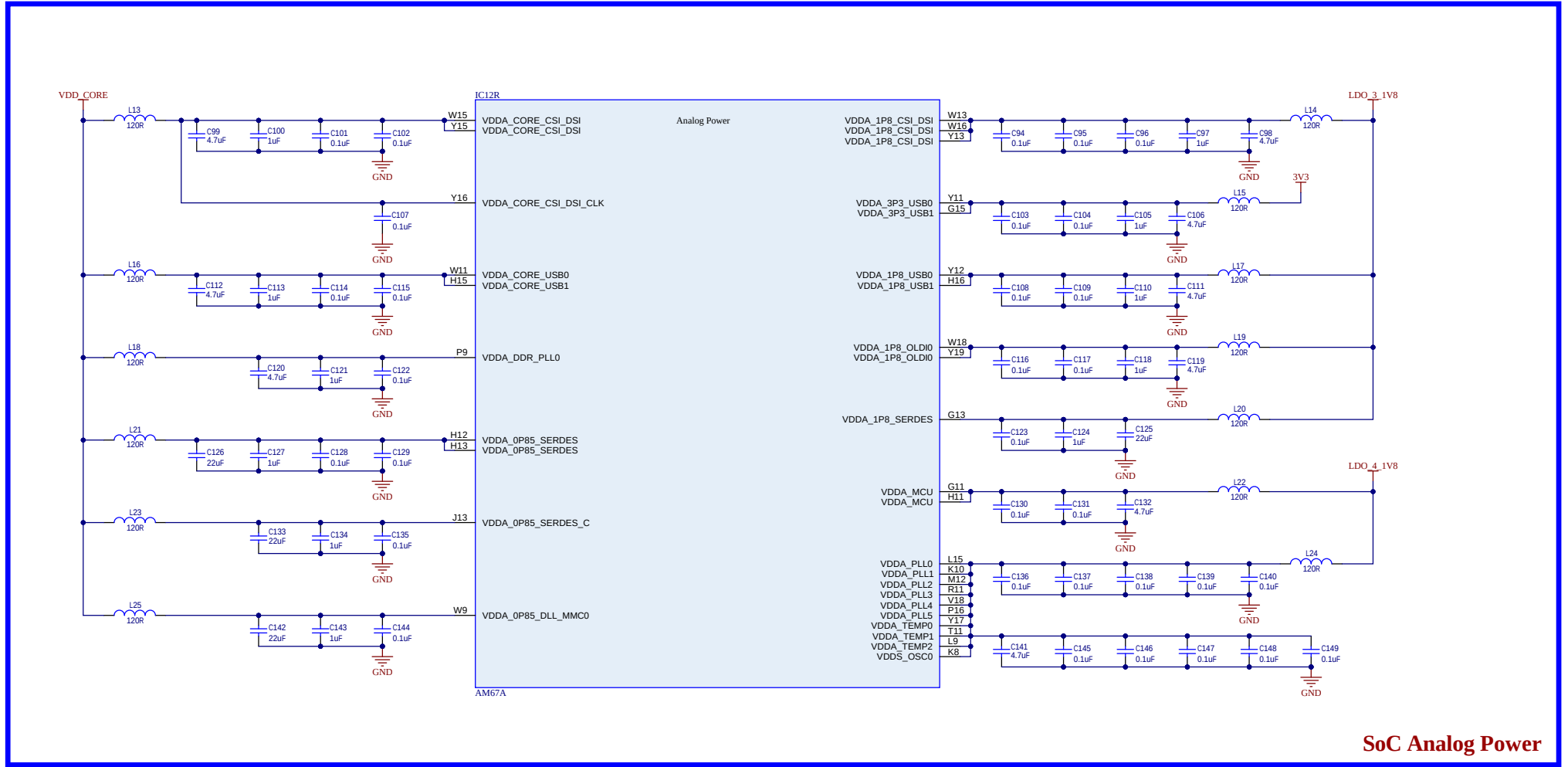


2.5 Volt Output

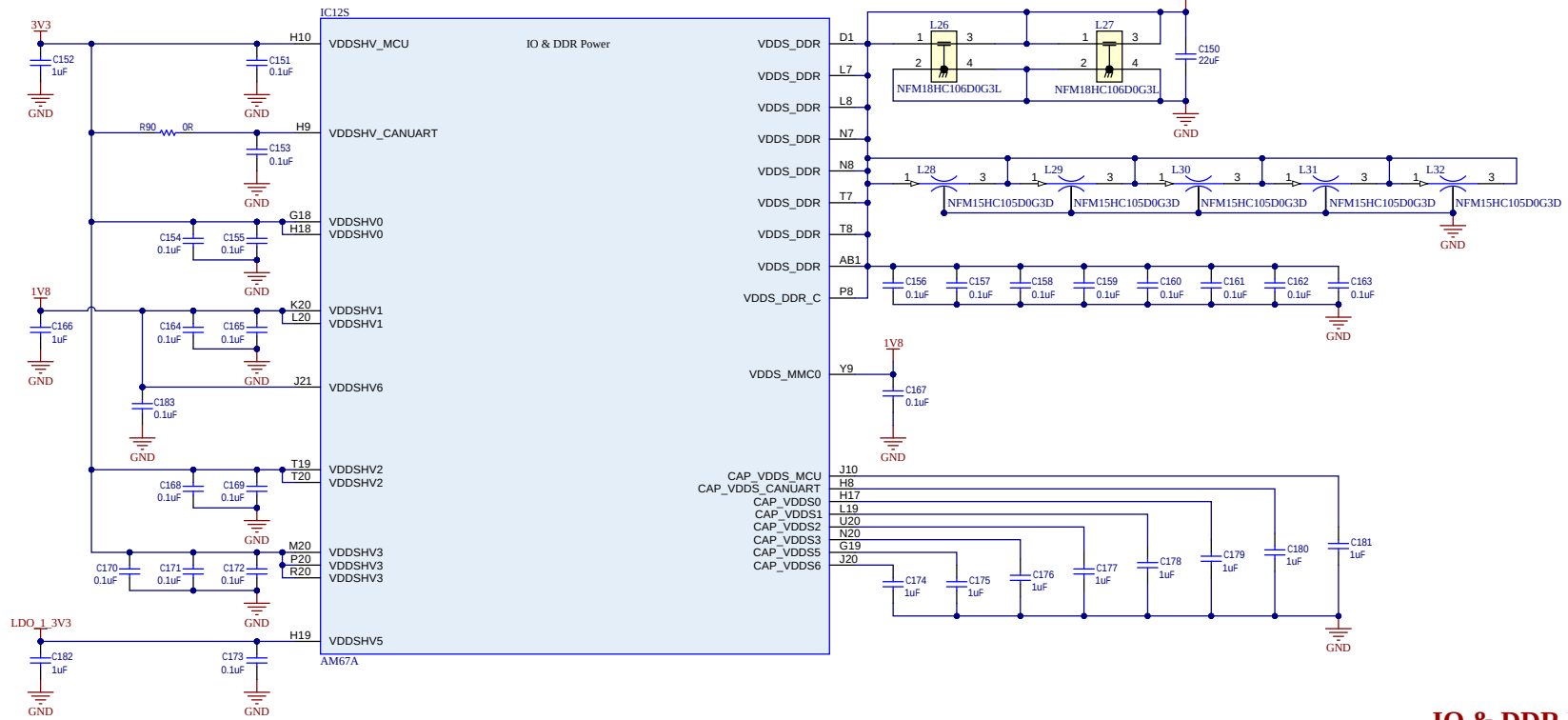


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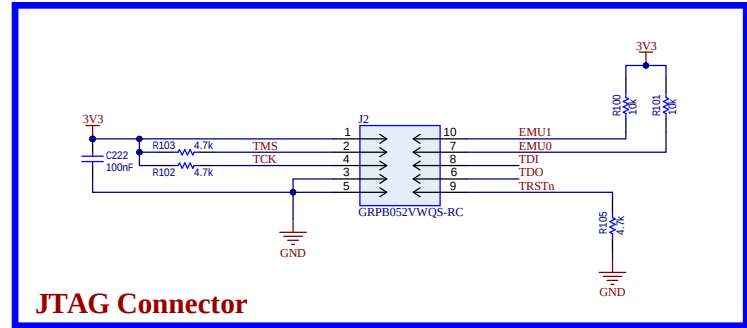
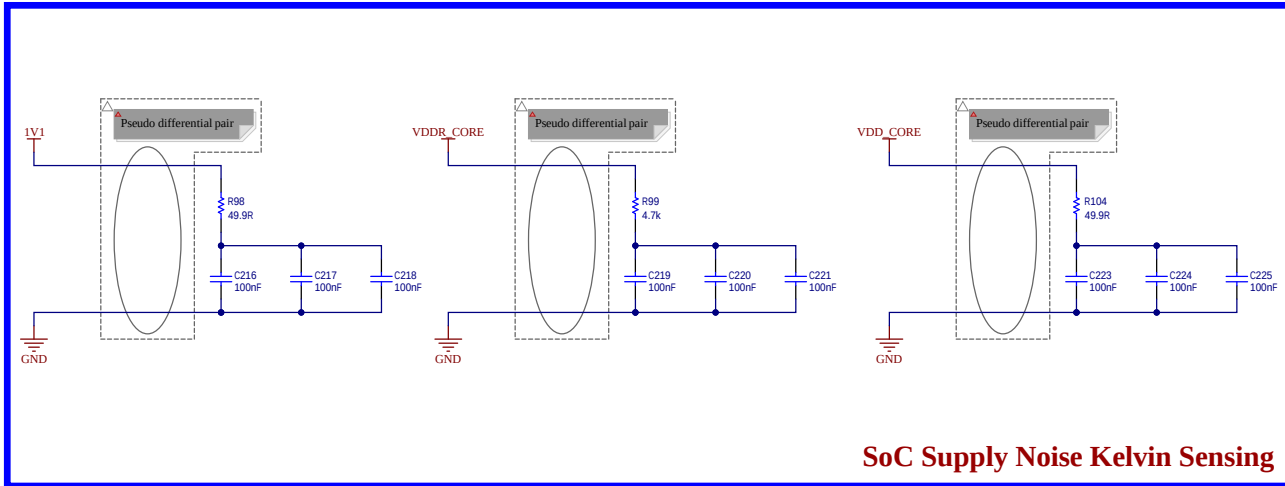
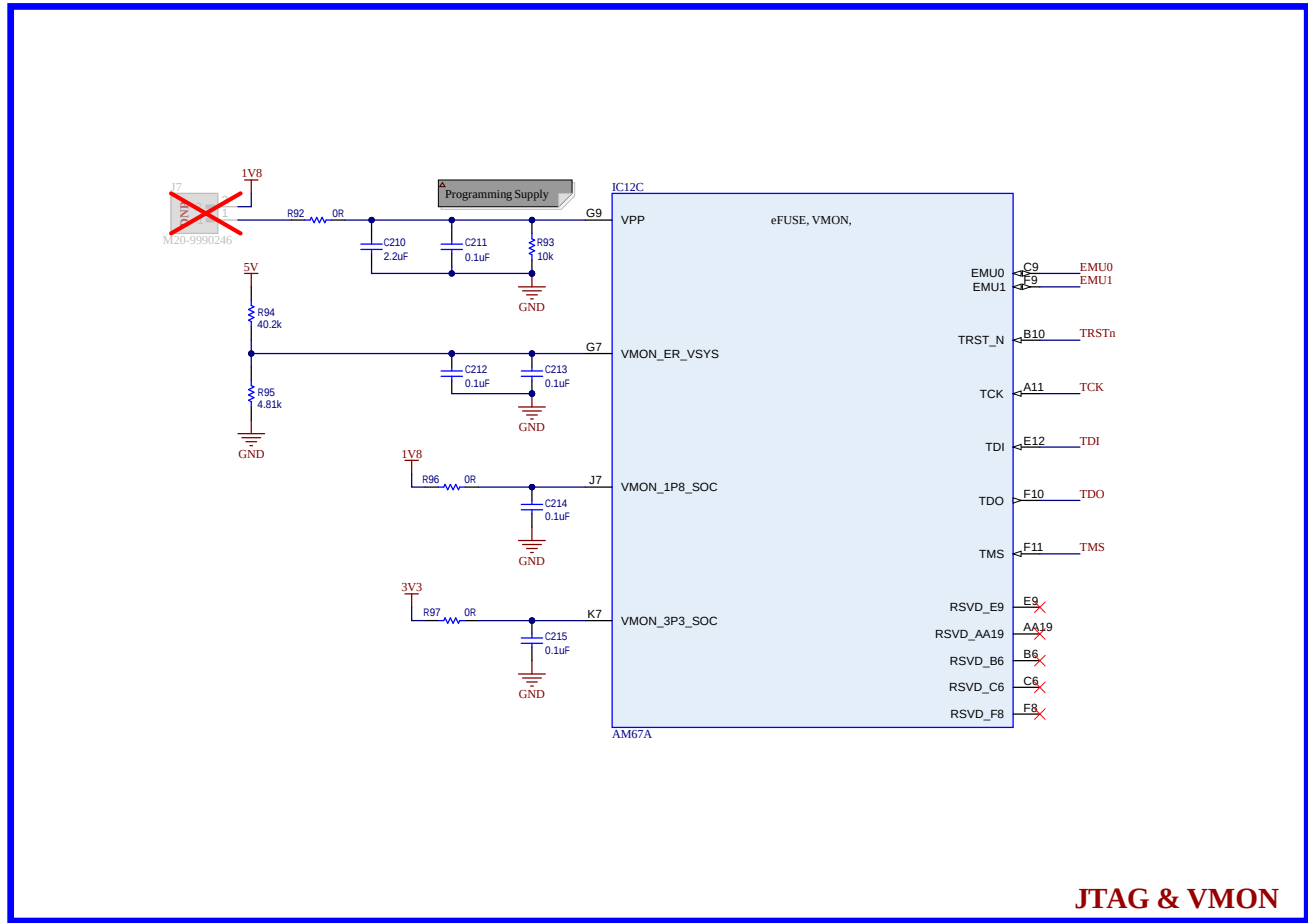
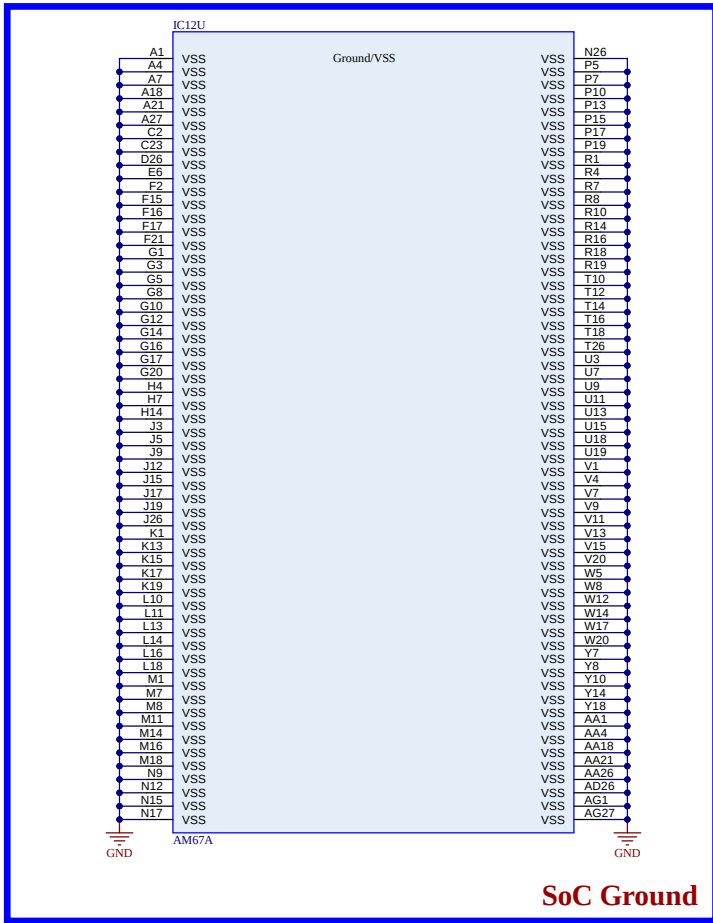
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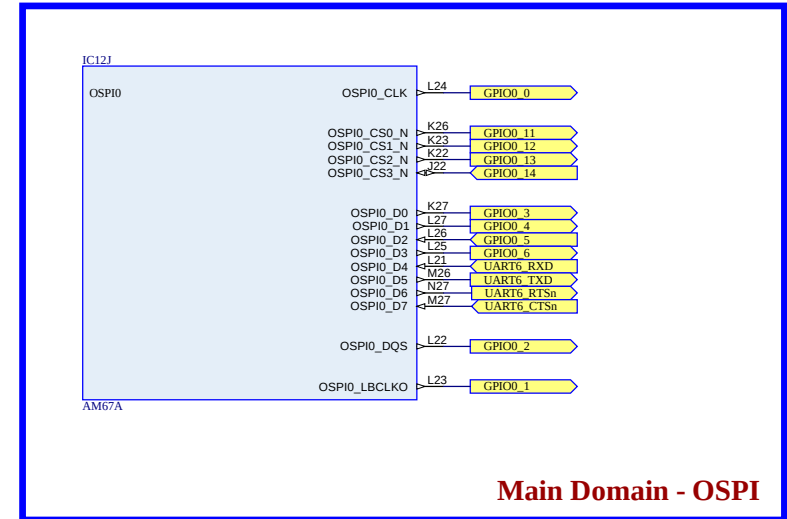
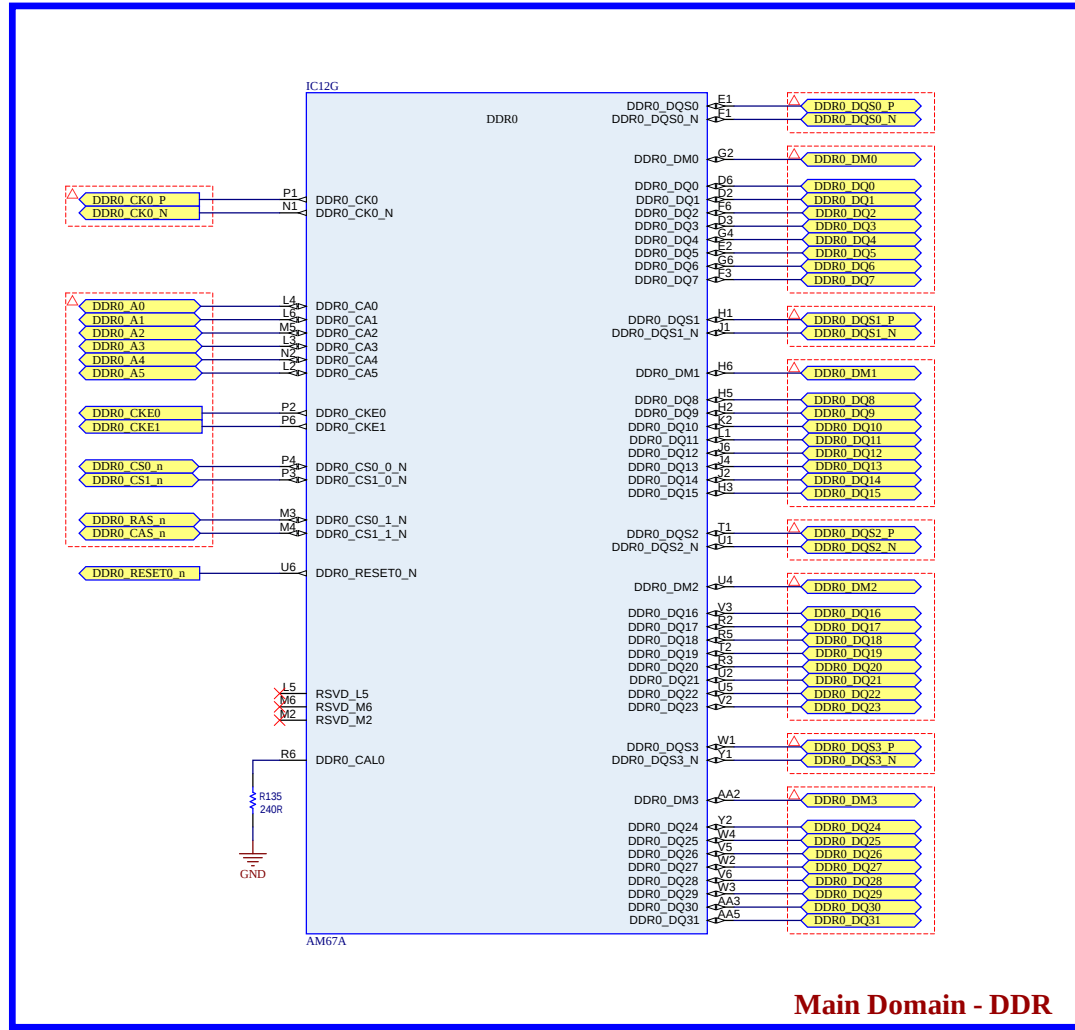
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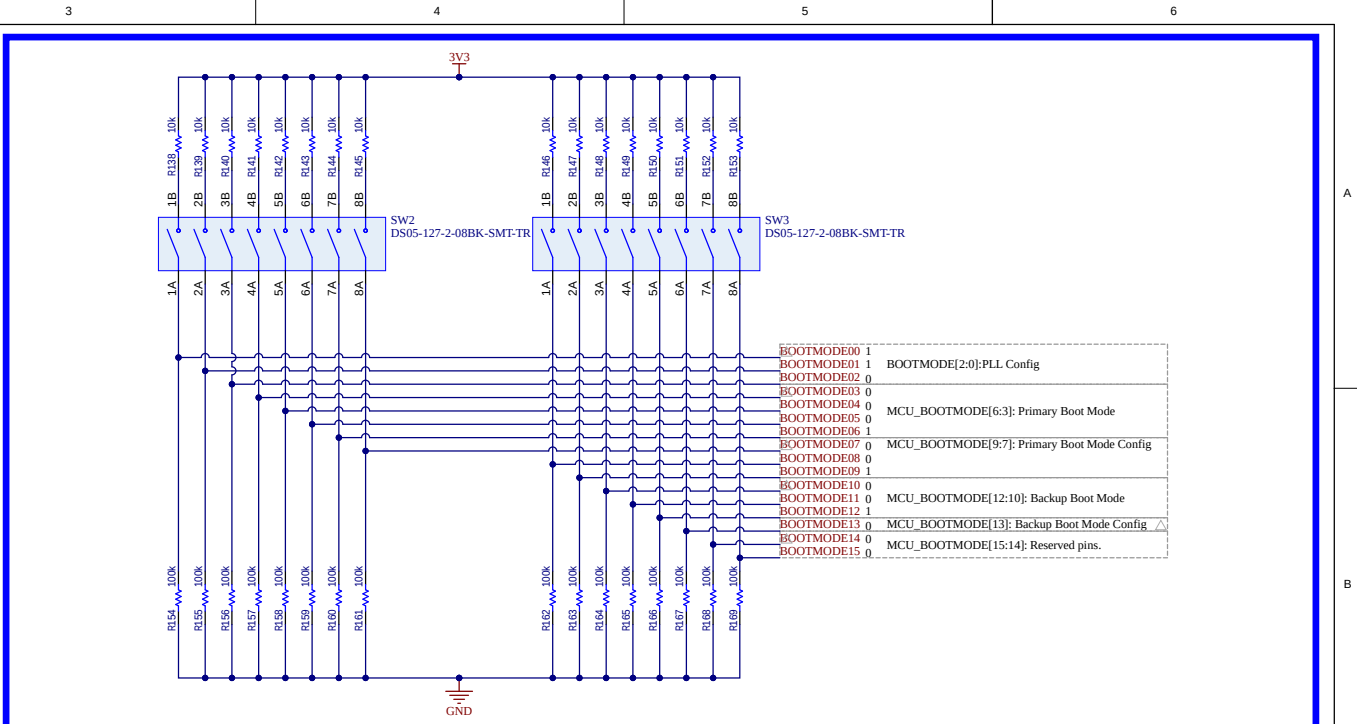
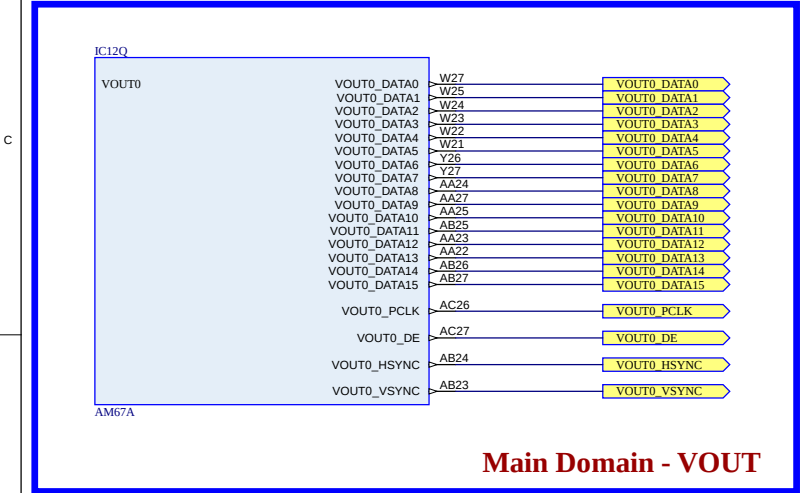
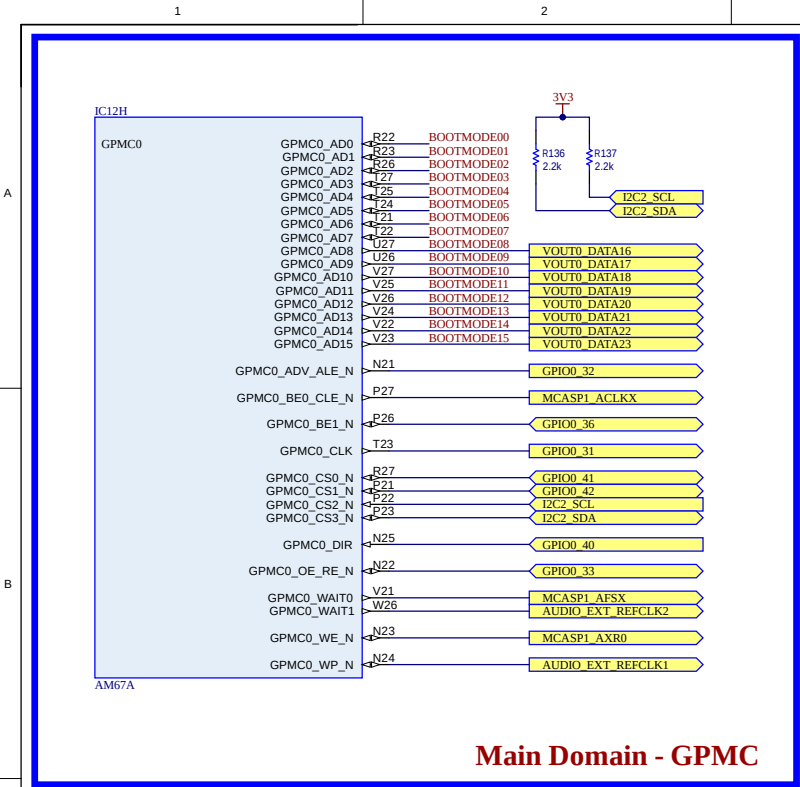


Table 5-2. BOOTMODE Pin Mapping

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved	Reserved	Backup Boot Mode Config	Backup Boot Mode			Primary Boot Mode Config			Primary Boot Mode			PLL Config			

Table 5-4. Primary Boot Mode Selection

Primary Boot Mode Config						Primary Boot Mode	
B9	B8	B7	B6	B5	B4	B3	
Reserved	Read Mode 2	Read Mode 1	0	0	0	0	Serial NAND
Reserved	Reserved	Csel	0	0	0	1	QSPI
Reserved	Reserved	Csel	0	0	1	0	QSPI
Reserved	Mode	Csel	0	0	1	1	SPI

Table 5-4. Primary Boot Mode Selection (continued)

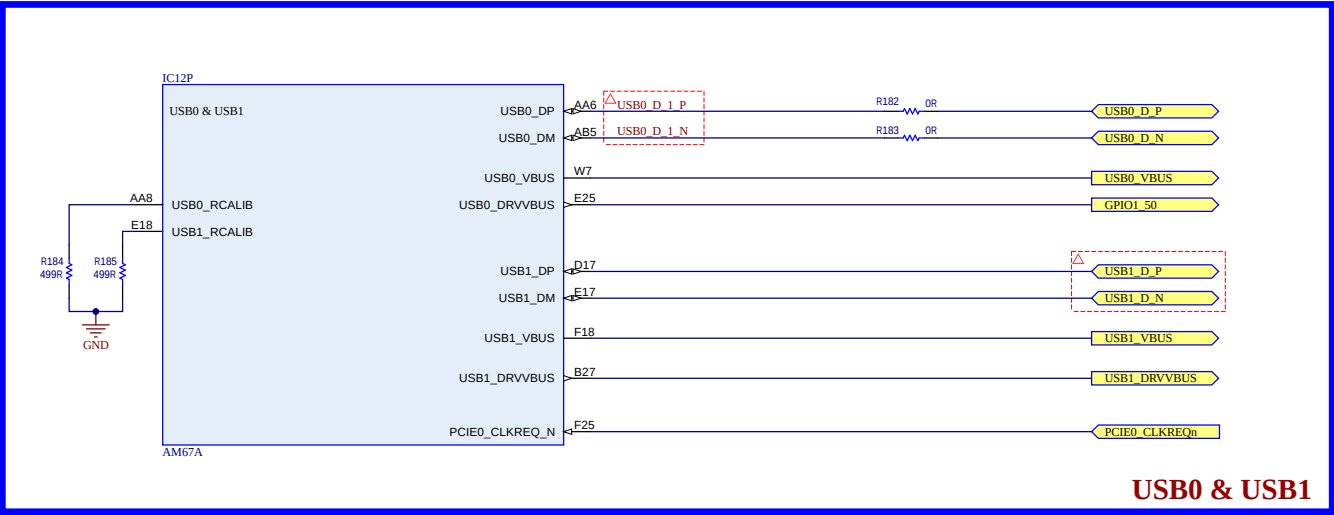
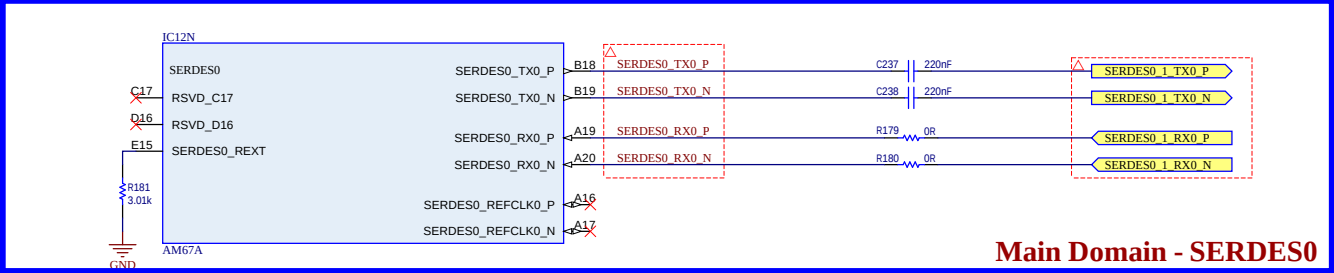
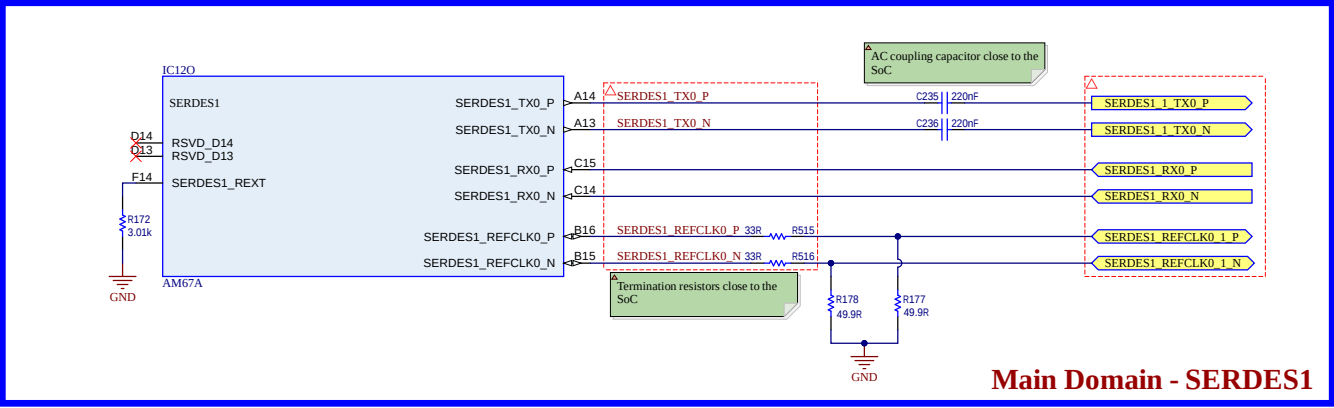
Primary Boot Mode Config						Primary Boot Mode	
B9	B8	B7	B6	B5	B4	B3	
0	0	Link Info	0	1	0	0	Ethernet RGMII
Clkout	Clk src	0	0	1	0	1	Ethernet RMII
Bus reset	Reserved	Addr	0	1	1	0	I2C
Reserved	Reserved	Reserved	0	1	1	1	UART
1	Reserved	Fa/raw	1	0	0	0	MMCSD Boot (SD Card Boot or eMMC Boot using UDA)
Reserved	Reserved	Reserved	1	0	0	1	eMMC Boot
0	Mode	Lane Swap	1	0	1	0	USB
Reserved	Reserved	Reserved	1	0	1	1	GPMC NAND
Reserved	Reserved	Reserved	1	1	0	0	GPMC NOR
Reserved	Reserved	Reserved	1	1	0	1	Fast-xSPI
SFPD	Read Cmd	Mode	1	1	1	0	xSPI
Reserved	ARM/Thumb	No/Dev	1	1	1	1	No-boot/Dev boot

Table 5-5. Backup Mode Selection

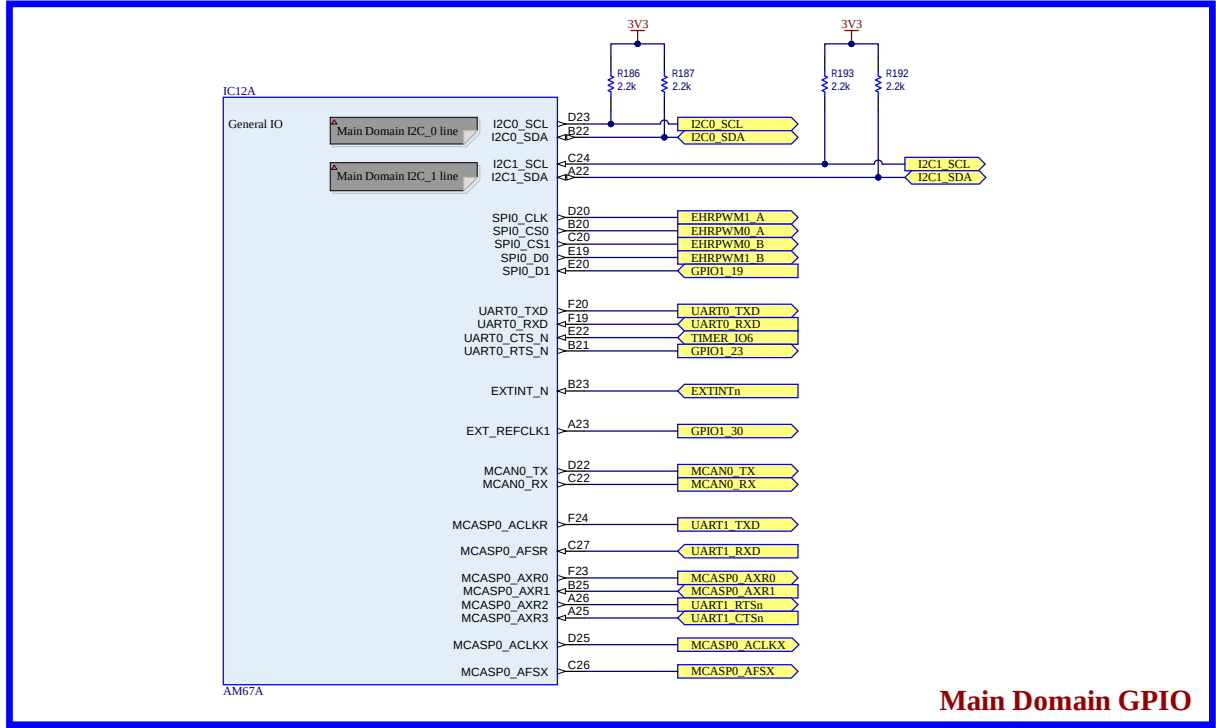
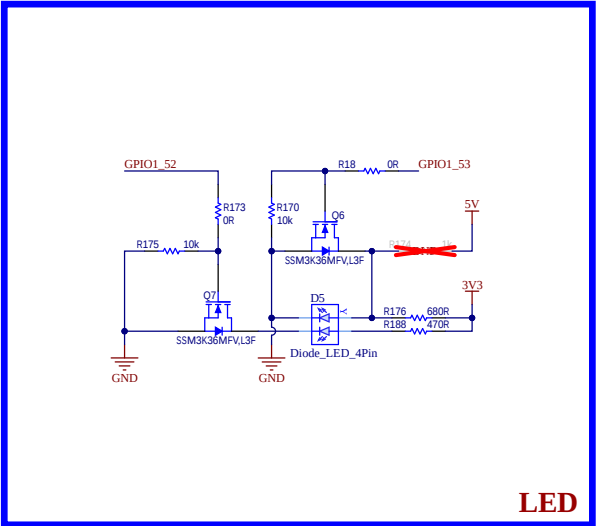
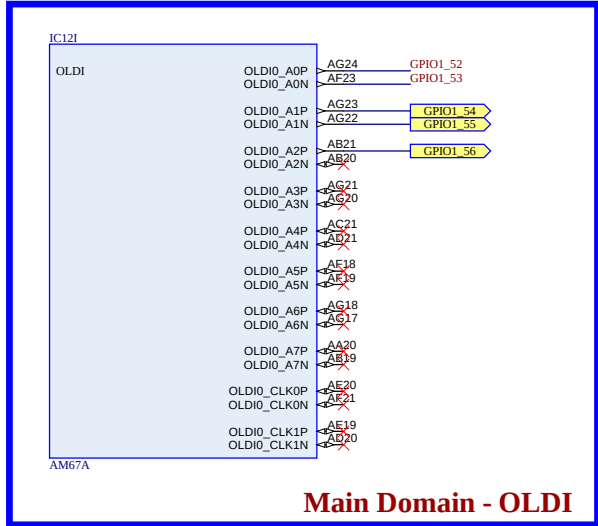
Backup Boot Config	Backup Boot Mode Selection				Backup Boot Mode Selected
B13	B12	B11	B10		
Reserved	0	0	0		None
Mode	0	0	1		USB
Reserved	0	1	0		Reserved
Reserved	0	1	1		UART
IF	1	0	0		Ethernet
Port	1	0	1		MMCSD Boot (SD Card Boot or eMMC Boot)
Reserved	1	1	0		SPI
Reserved	1	1	1		I2C



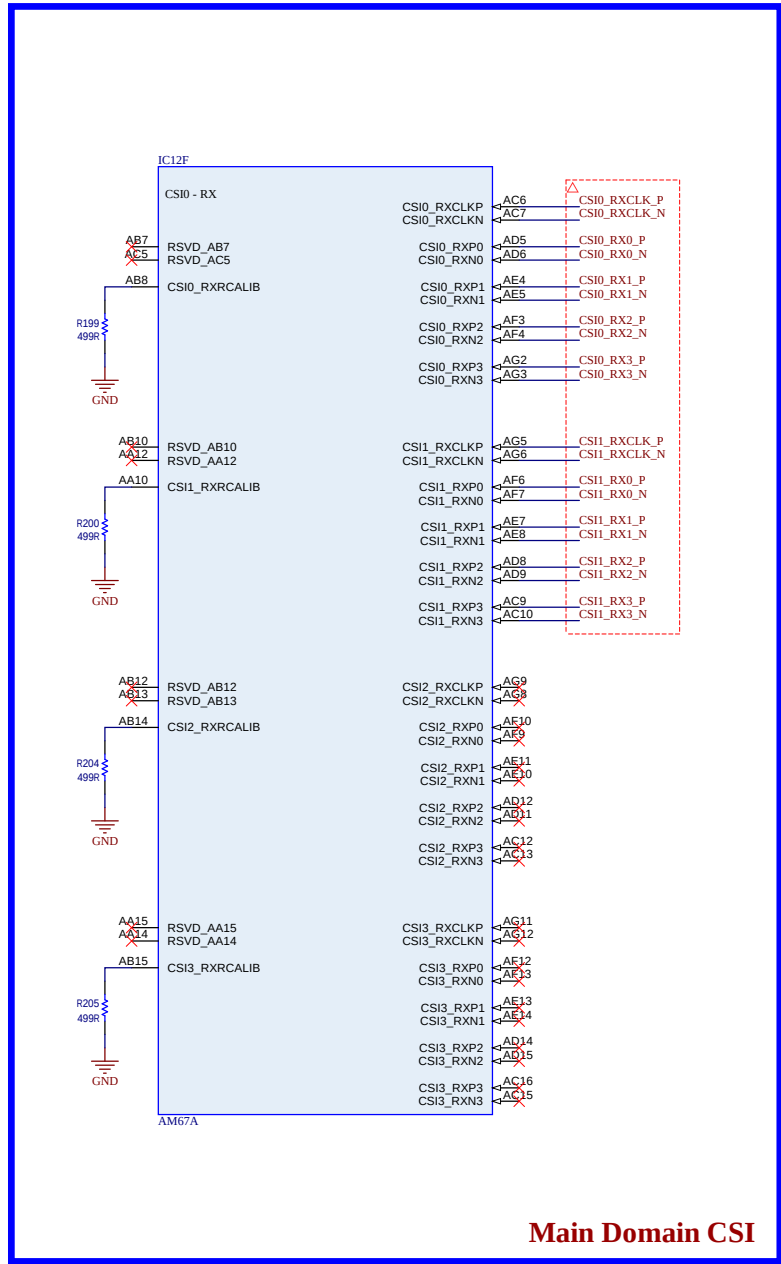
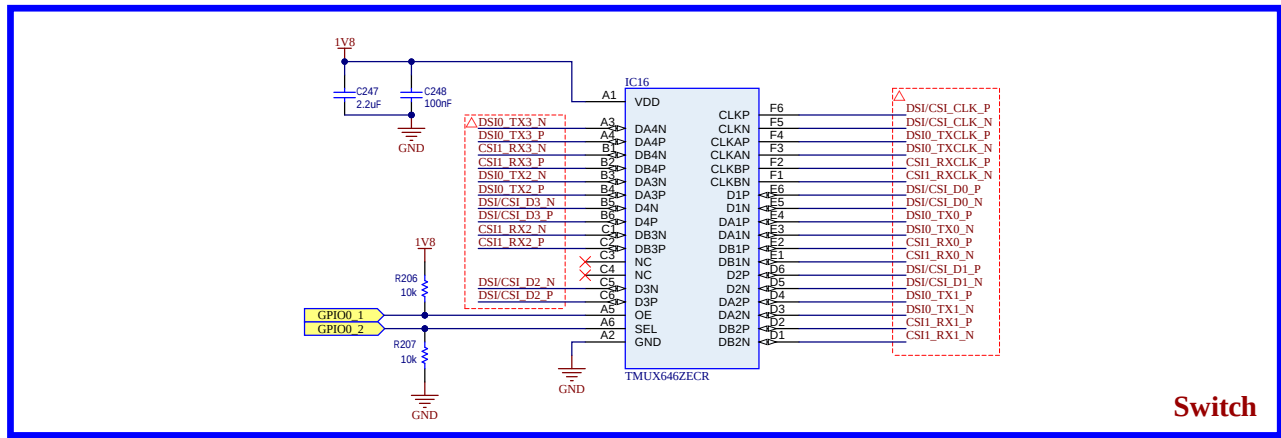
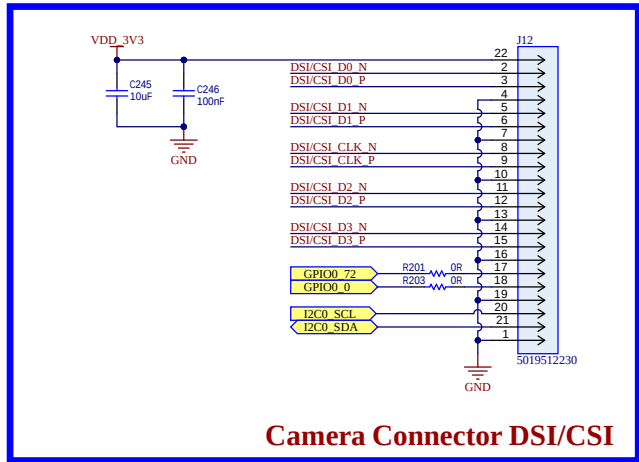
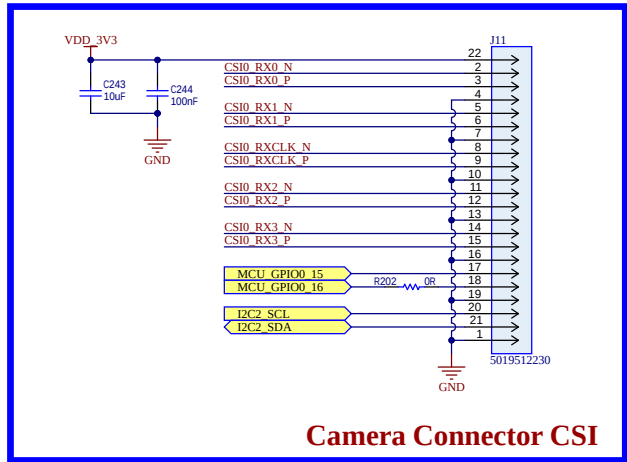
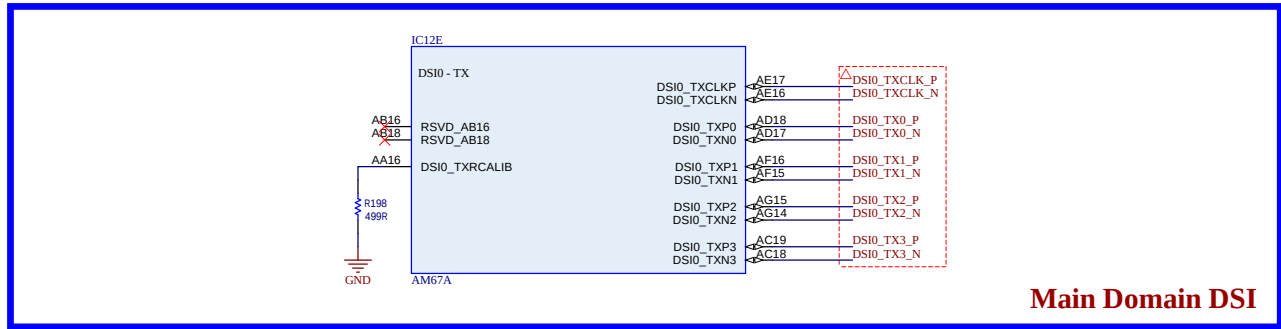
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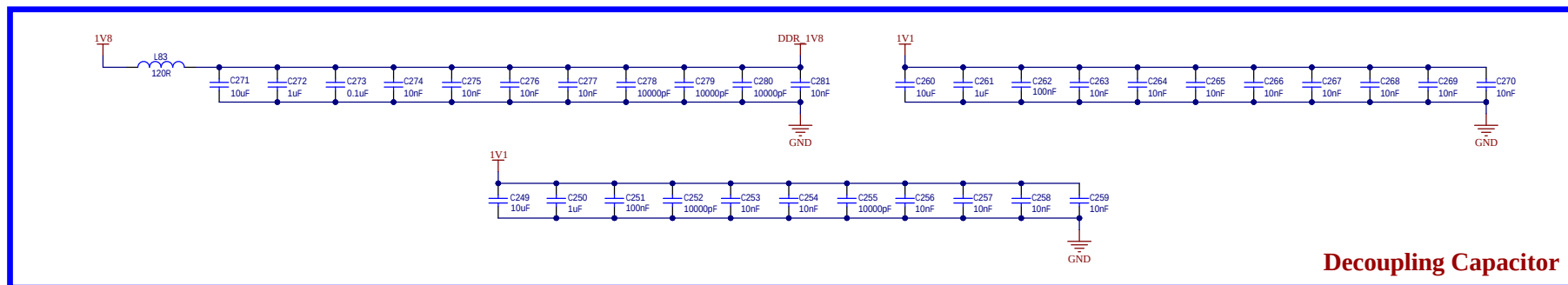
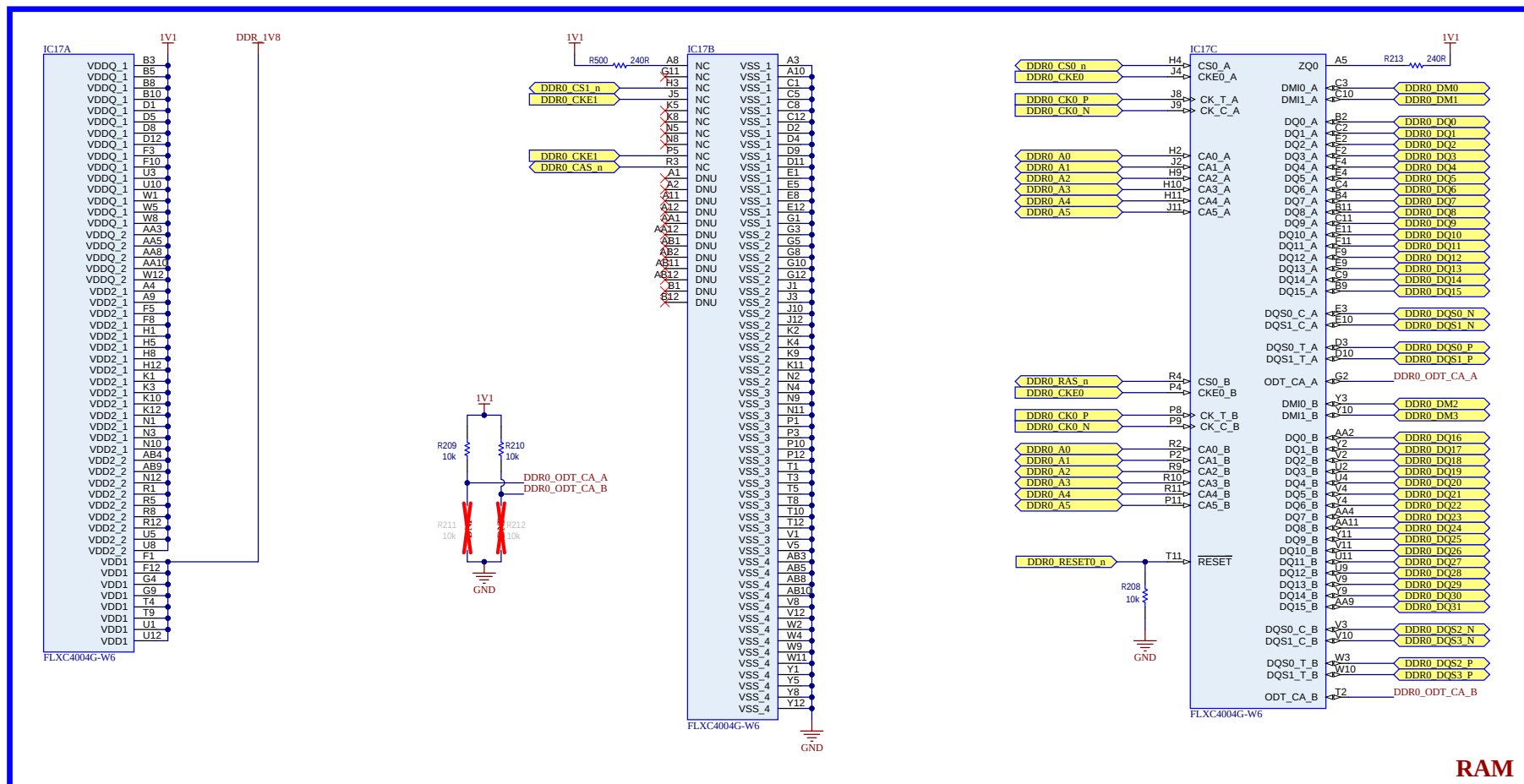


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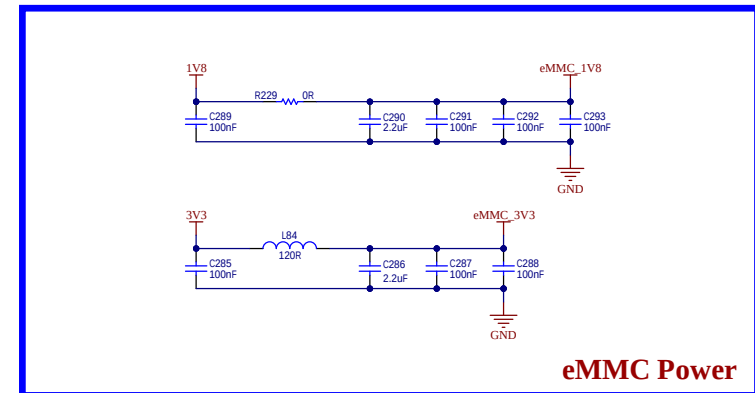
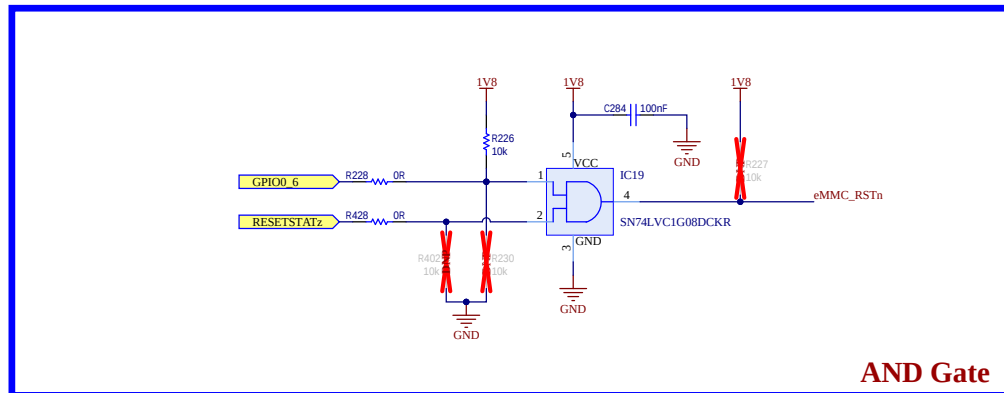
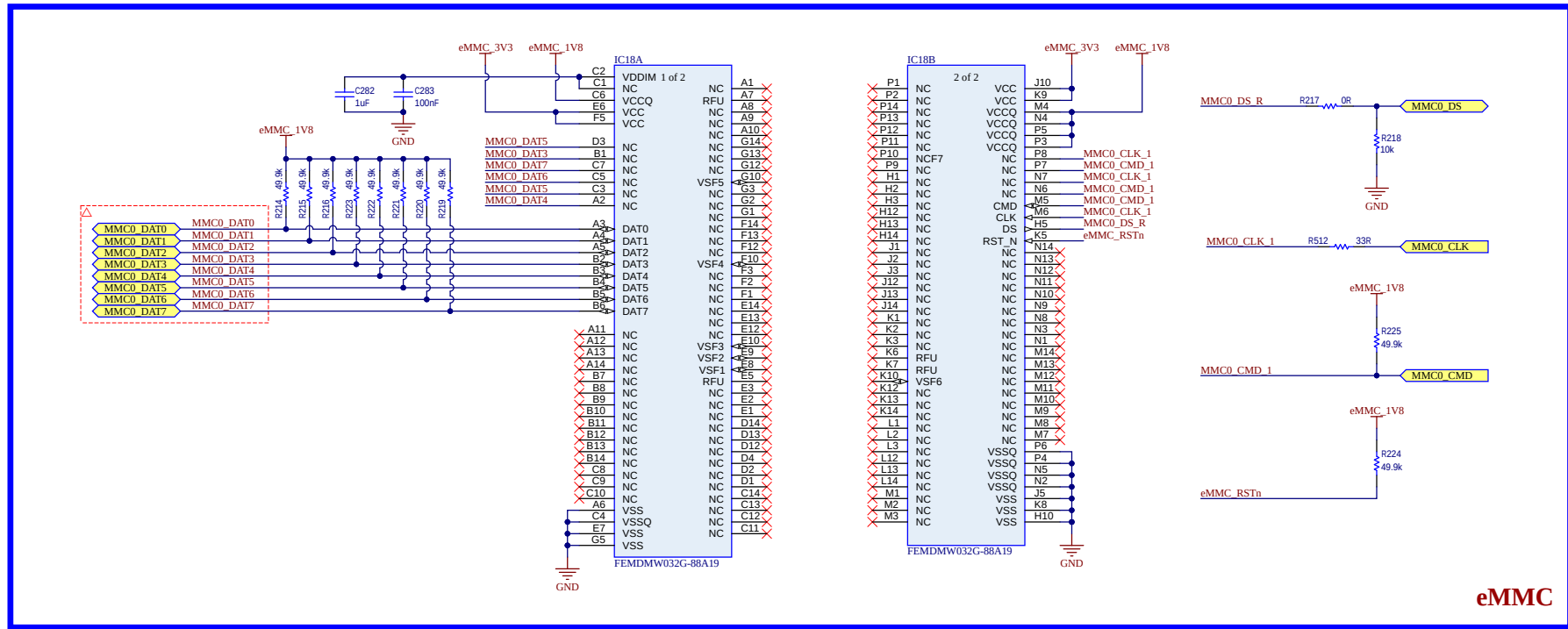
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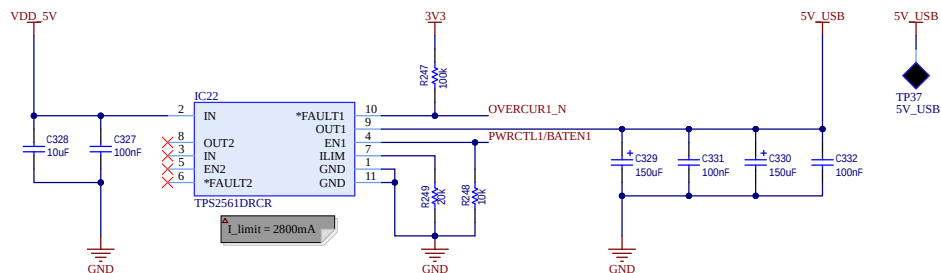
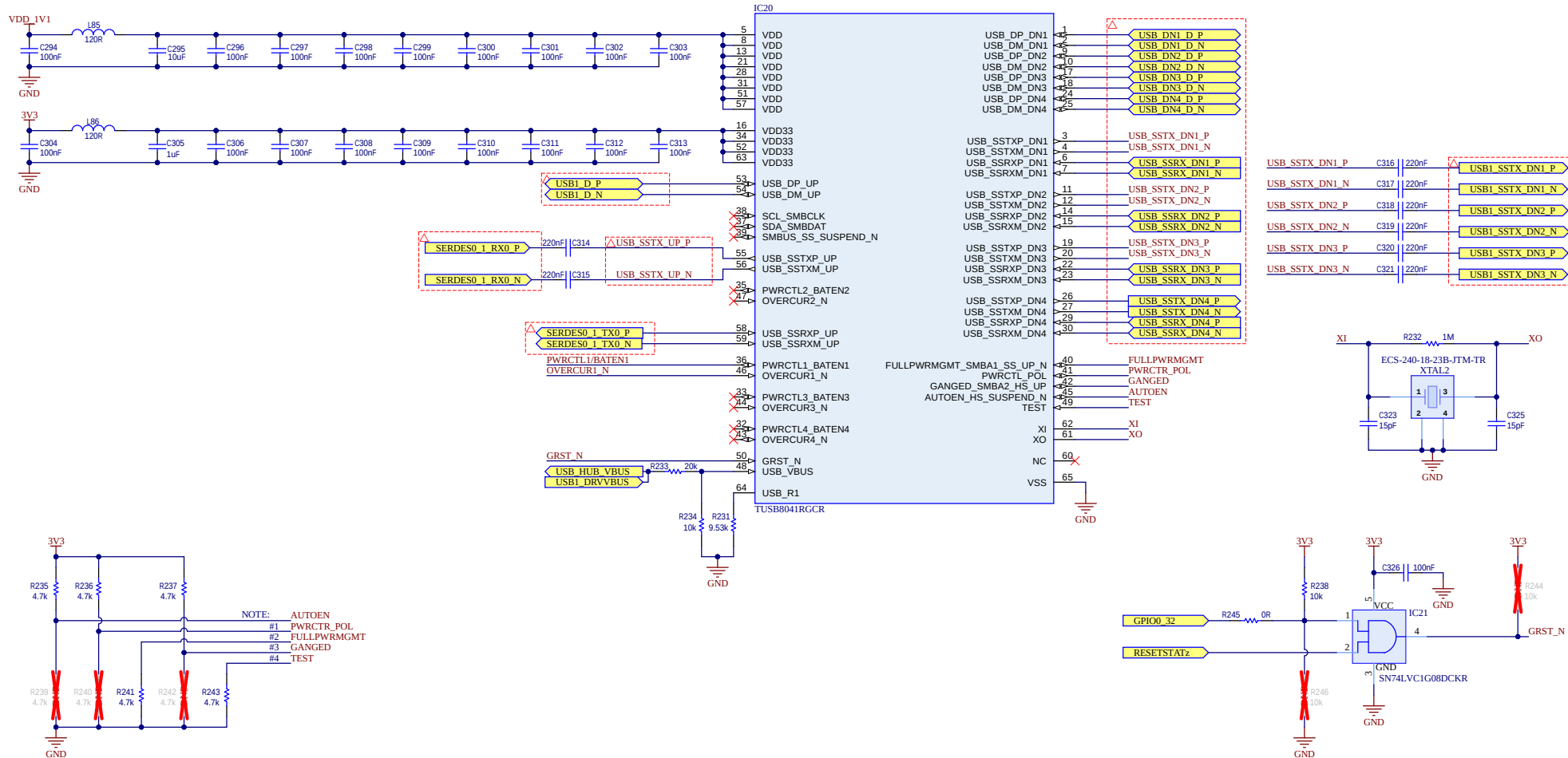




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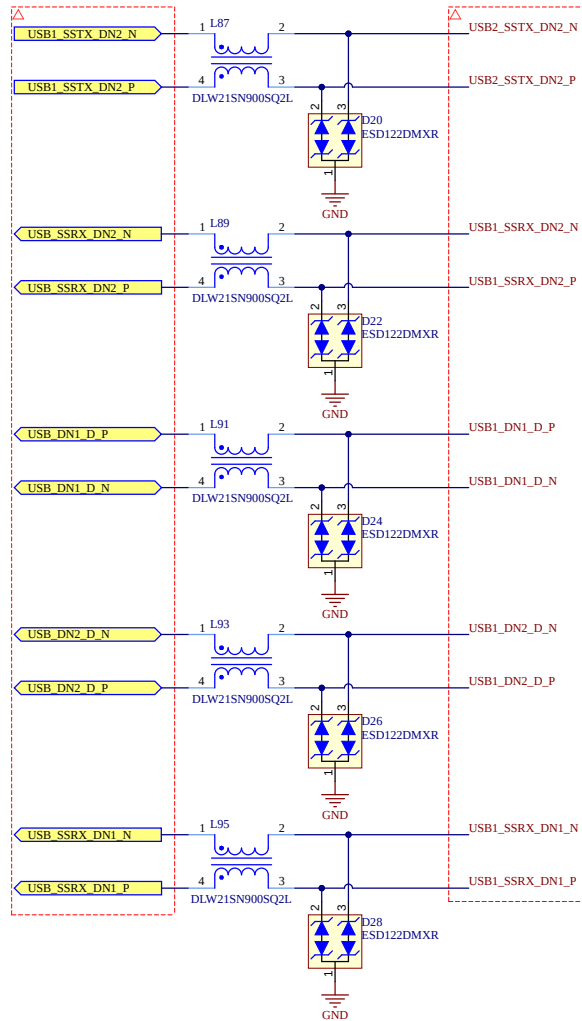
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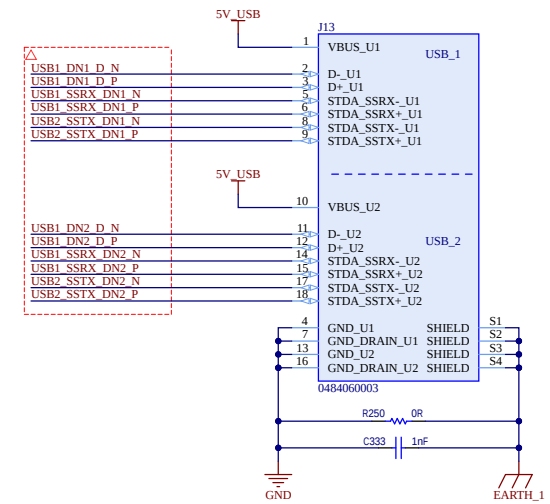


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USB Data Filter

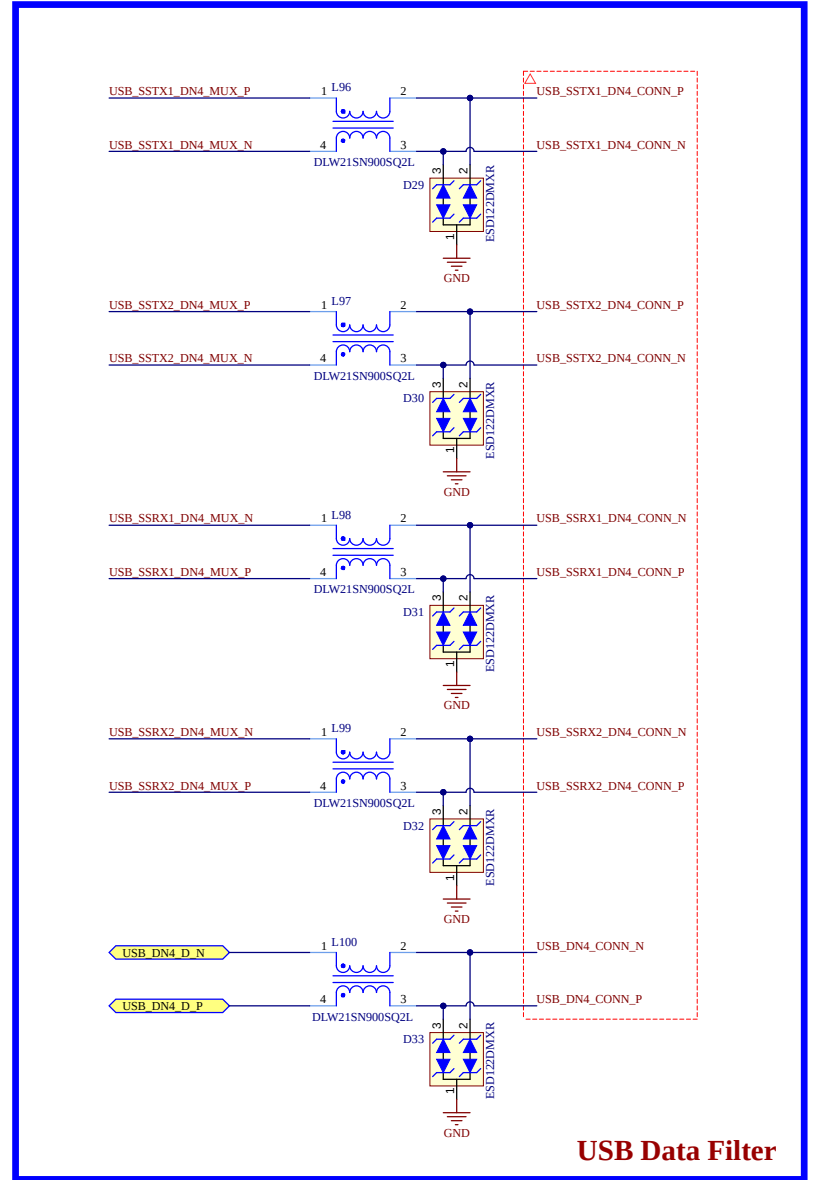
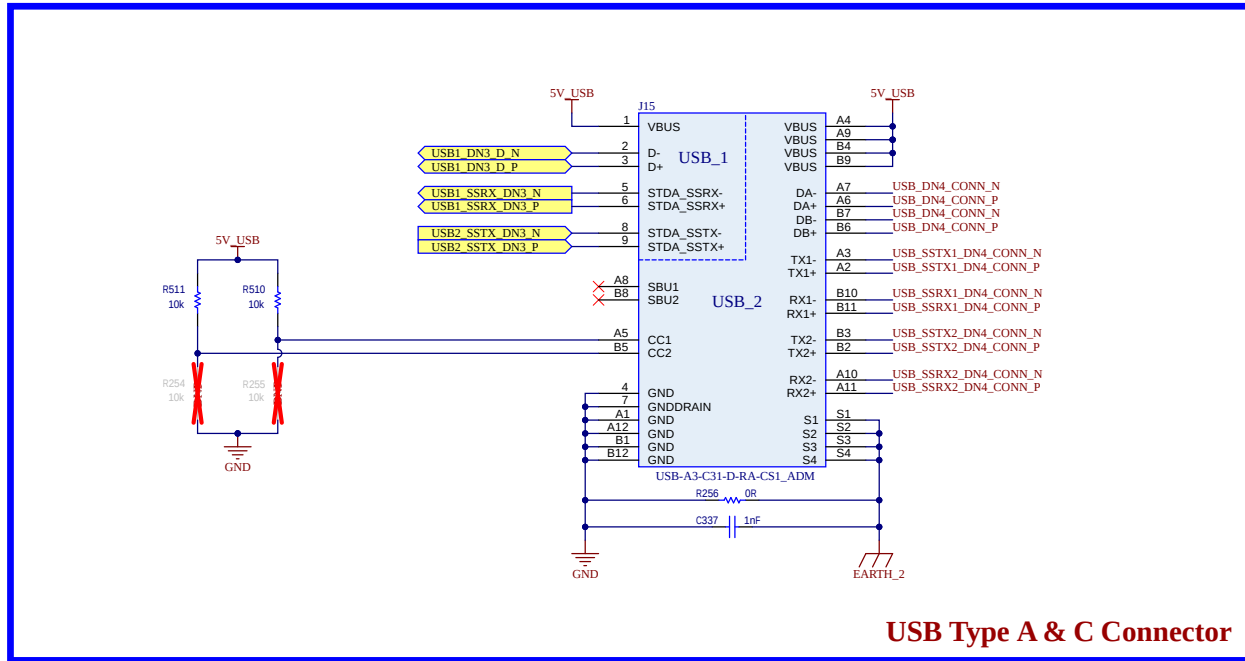
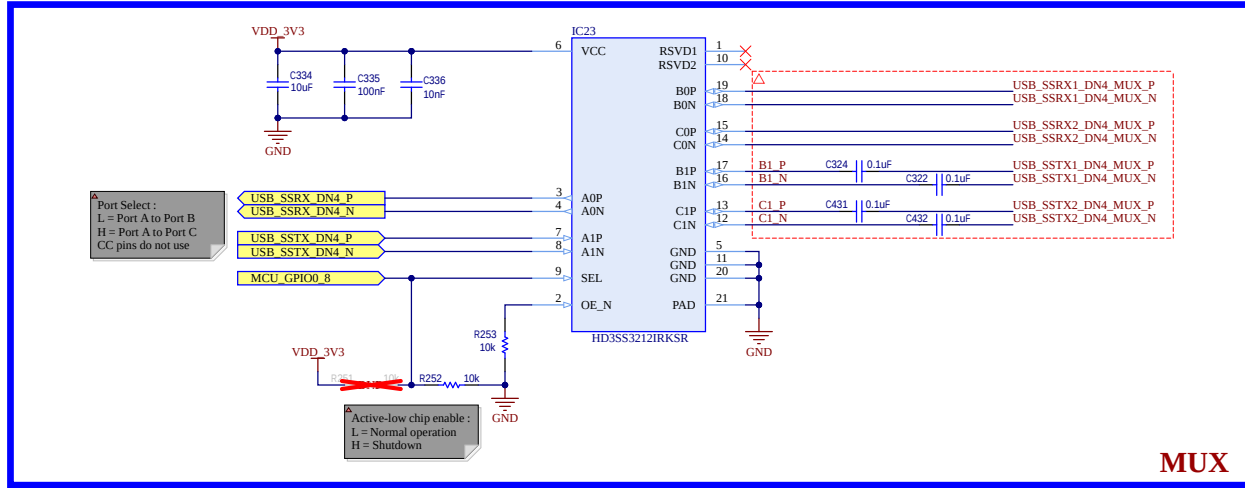


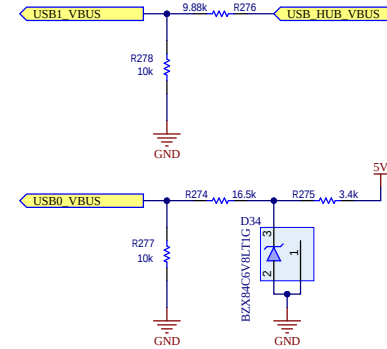
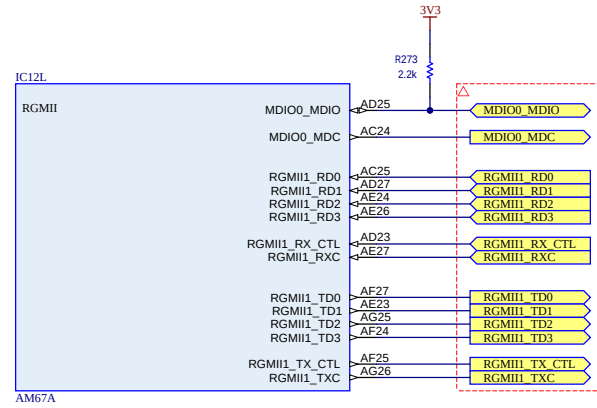
USB TYPE A Connector



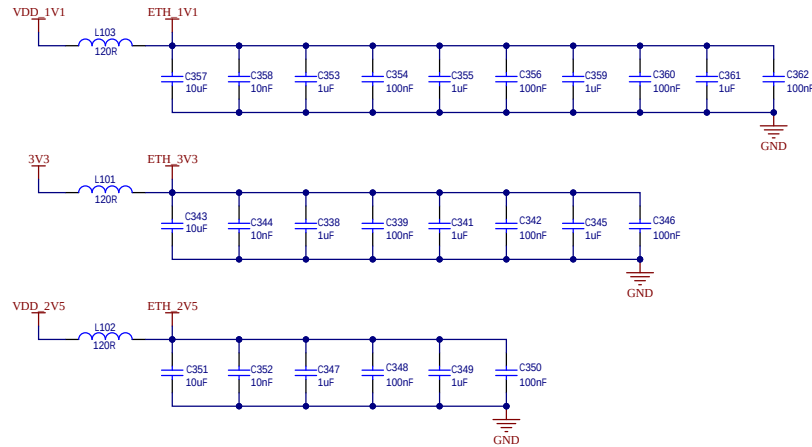
T3 Gemstone

Proje / Project:	T3_Gemstone
Sayfa Adı / Page Name:	USB_Connector.SchDoc
Revizyon / Revision:	V0.2
Sayfa / Page:	24 / 33
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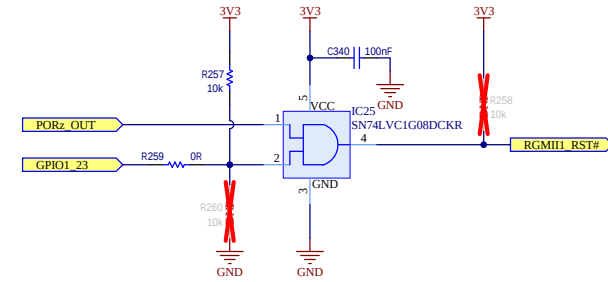




Main Domain RGMII



Ethernet Decoupling Capacitors



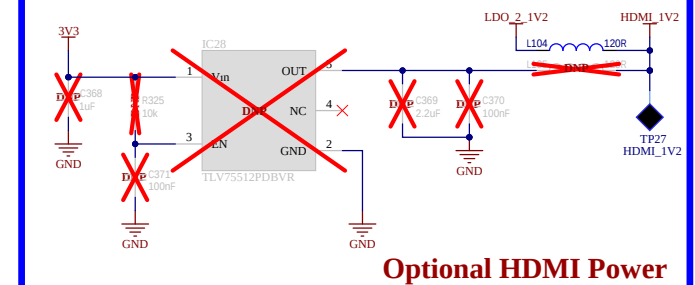
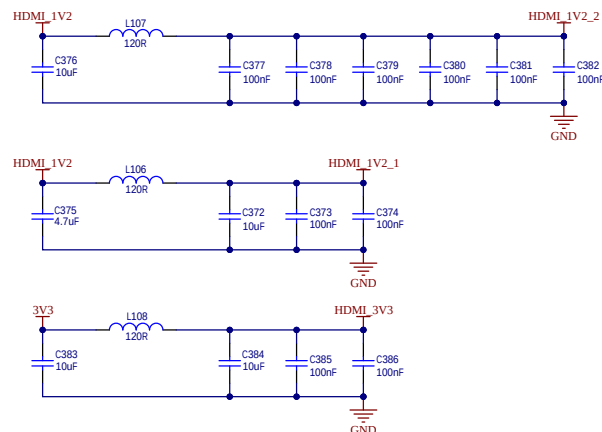
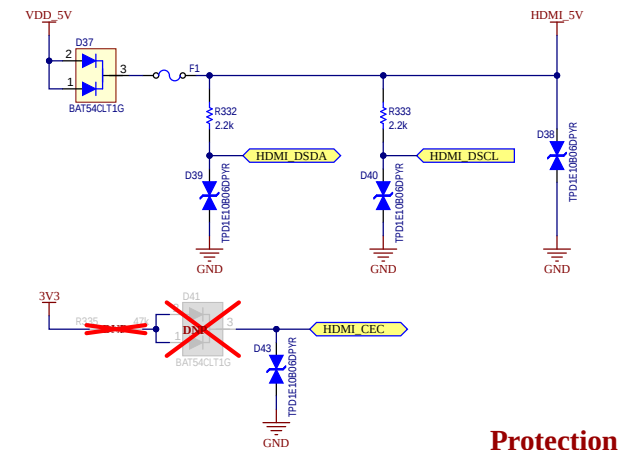
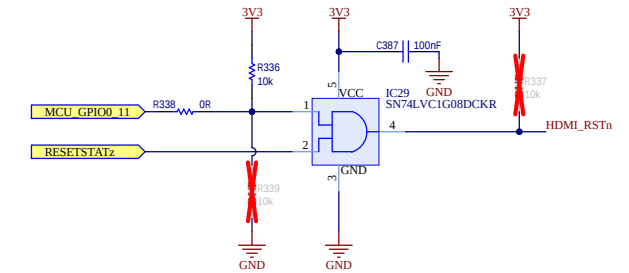
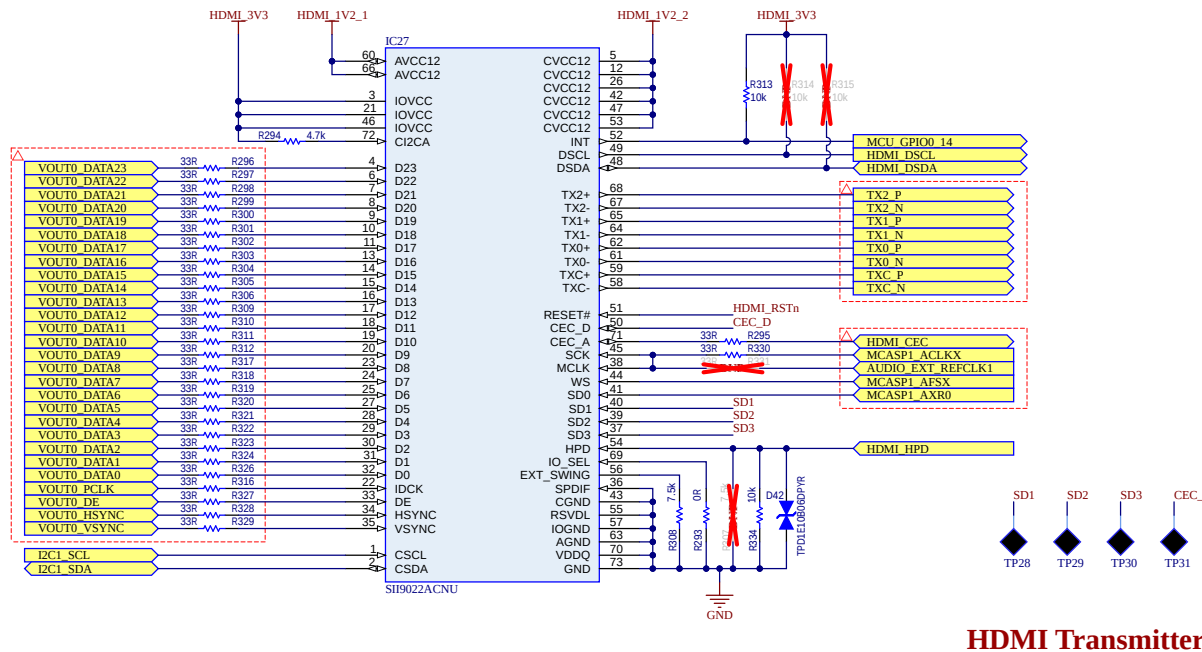
Ethernet AND Gate



Proje / Project:	T3_Gemstone
Sayfa Adı / Page Name:	SOC_RGMI - Ethernet_1.SchDoc
Revizyon / Revision:	V0.2
Sayfa / Page:	26 / 33
Firma / Organization:	T3 VAKFI
Tarih / Date:	15.07.2025

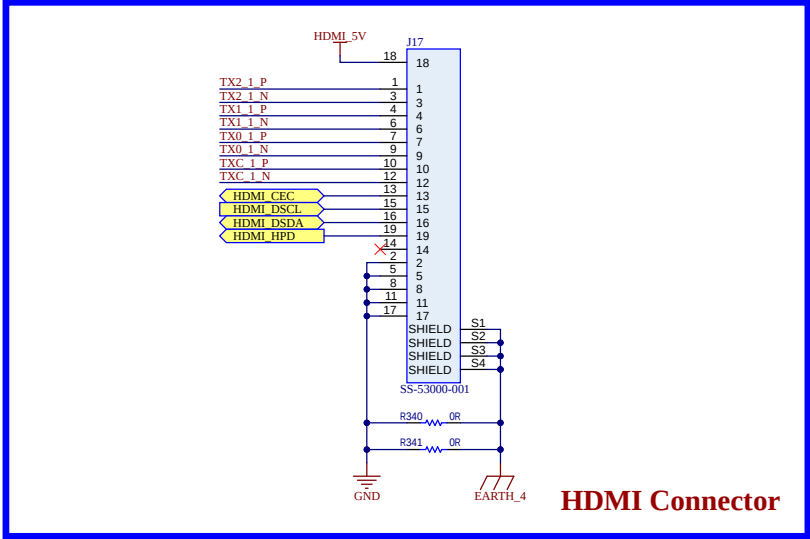
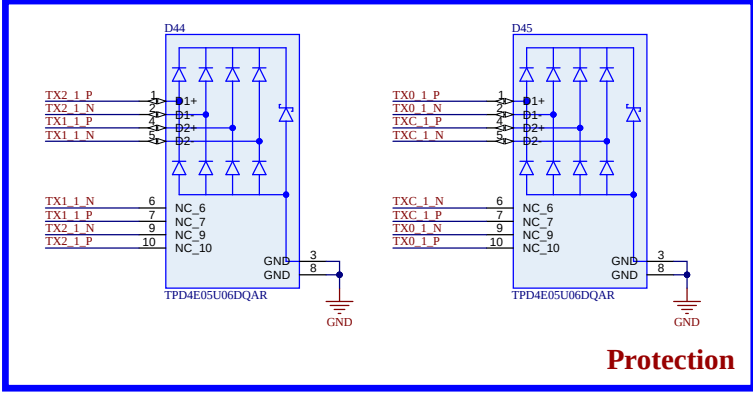
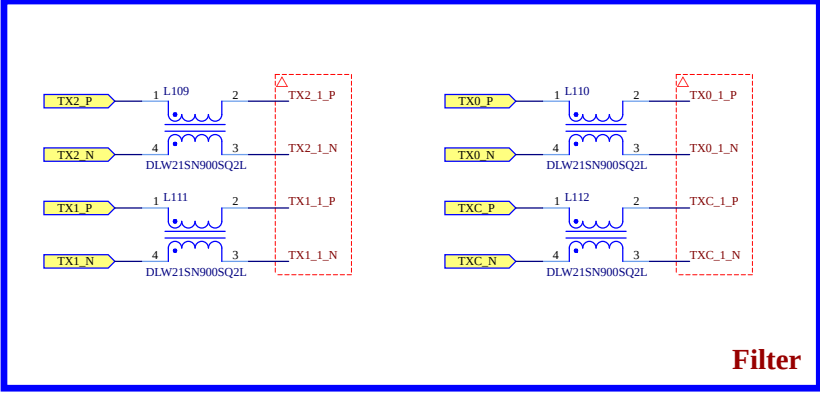


Proje / Project:	T3_Gemstone
Sayfa Adı / Page Name:	Ethernet_2.SchDoc
Revizyon / Revision:	V0.2
Sayfa / Page:	27 / 33
Firma / Organization:	T3 VAKFI
Tarih / Date	15.07.2025

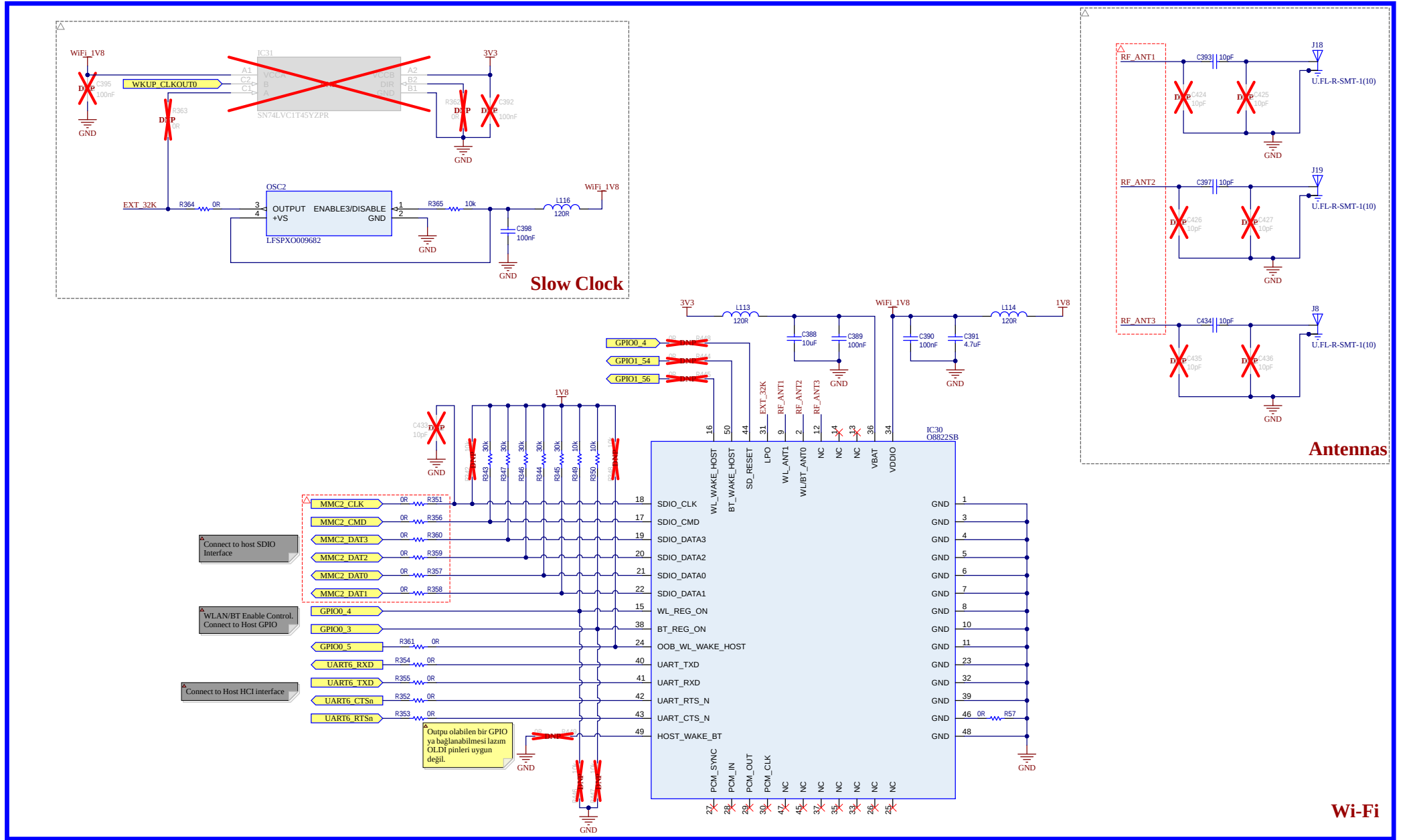


T3 Gemstone

Proje / Project:	T3_Gemstone
Sayfa Adı / Page Name:	HDMI_Transmitter.SchDoc
Revizyon / Revision:	V0.2
Sayfa / Page:	28 / 33
Firma / Organization:	T3 VAKFI
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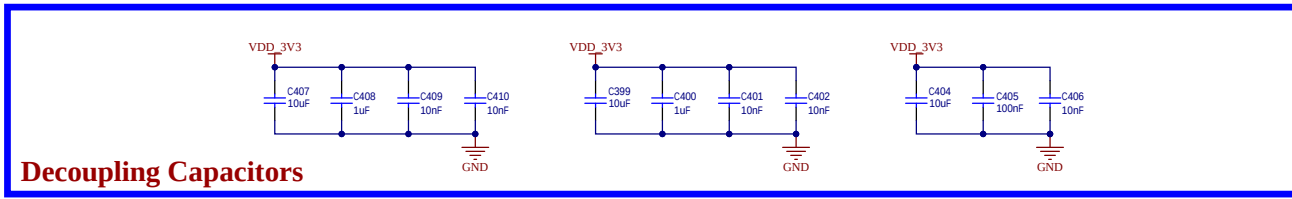
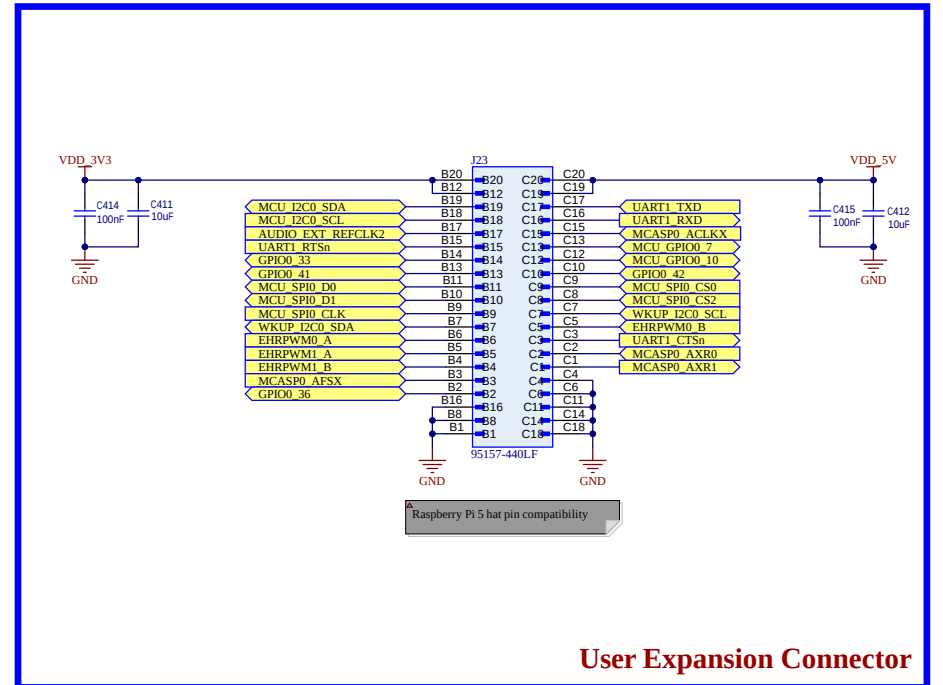
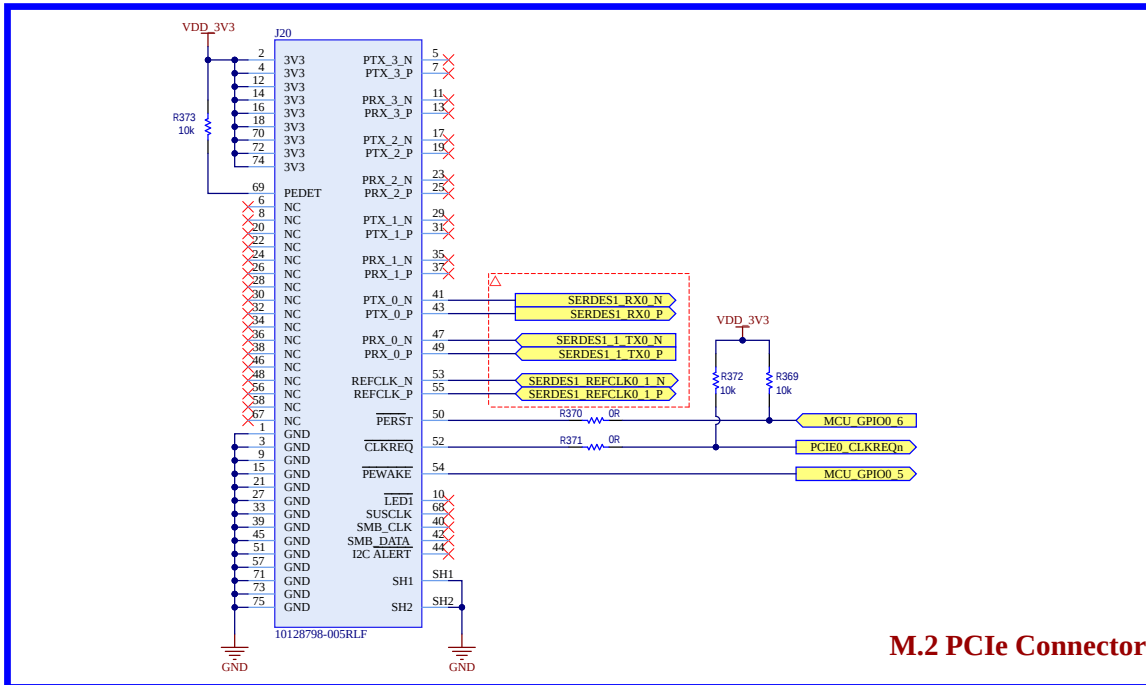
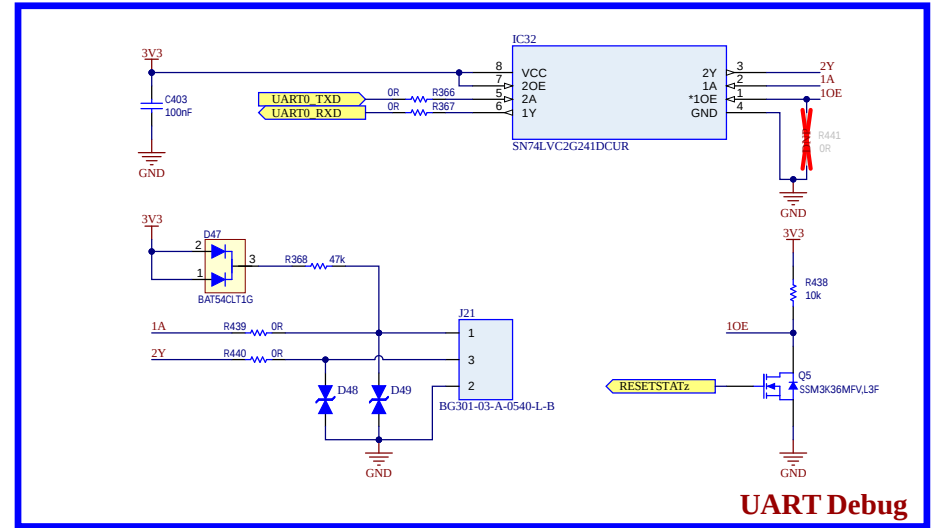
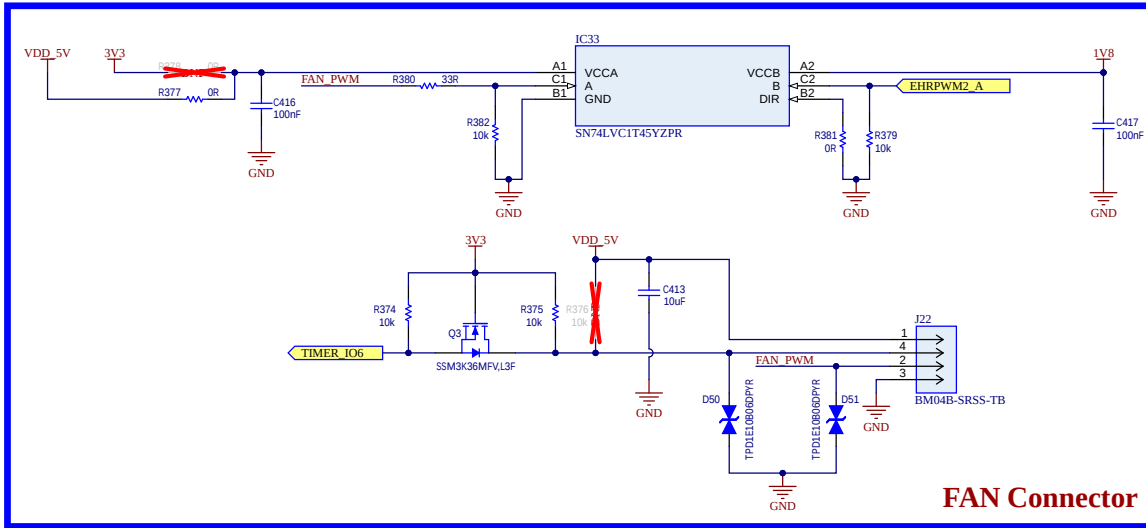


Proje / Project:	T3_Gemstone
Sayfa Adı / Page Name:	HDMI_Transmitter_1.SchDoc
Revizyon / Revision:	V0.2
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T3 Gemstone

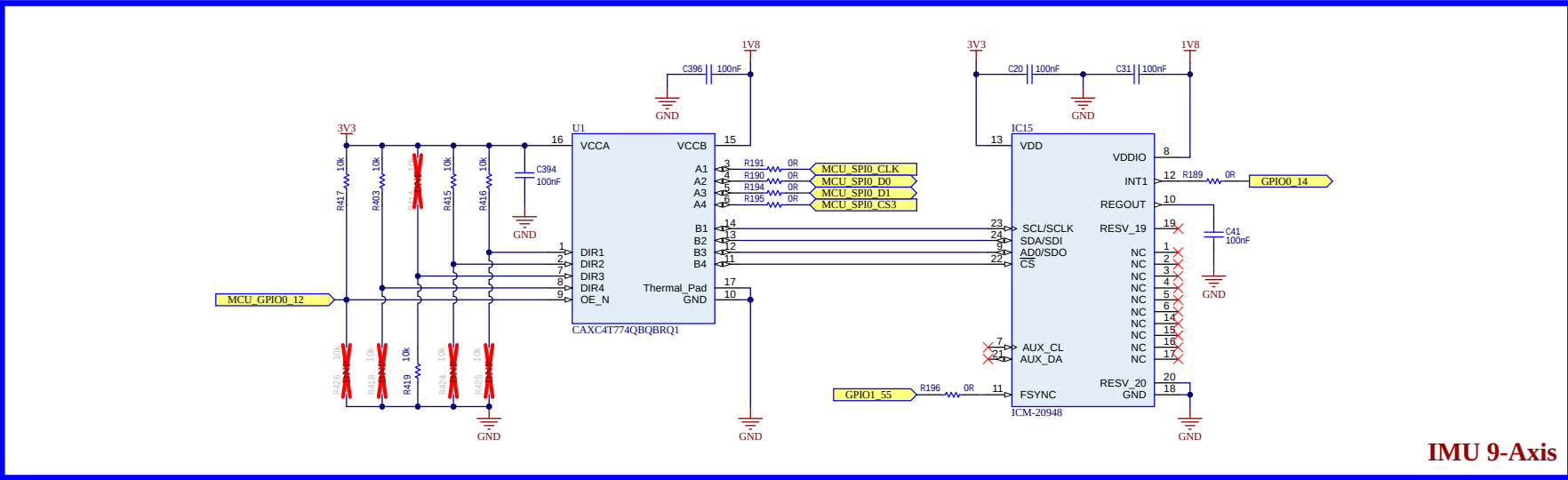
Proje / Project:	T3_Gemstone
Sayfa Adı / Page Name:	WiFi.SchDoc
Revizyon / Revision:	V0.2
Sayfa / Page:	30 / 33
Firma / Organization:	T3 VAKFI
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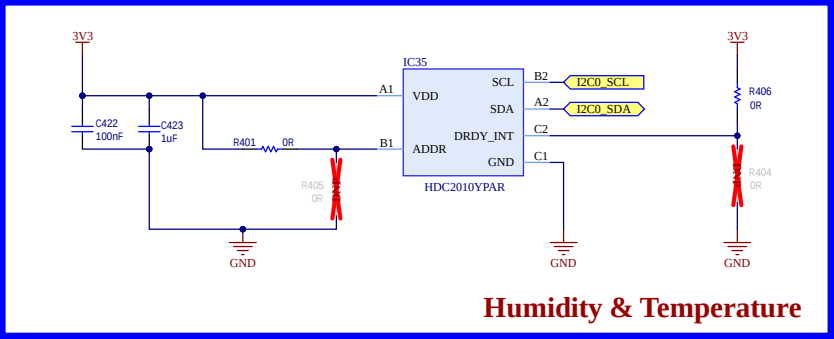
Proje / Project:	T3_Gemstone
Sayfa Adı / Page Name:	Connectors.SchDoc
Revizyon / Revision:	V0.2
Sayfa / Page:	31 / 33
Firma / Organization:	T3 VAKFI
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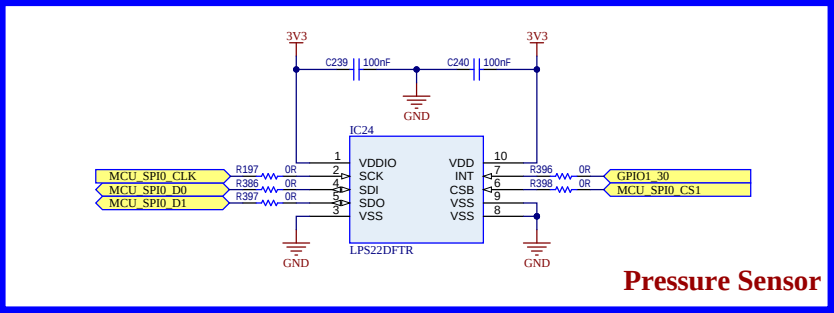
Proje / Project:	T3_Gemstone
Sayfa Adı / Page Name:	RTC - EEPROM.SchDoc
Revizyon / Revision:	V0.2
Sayfa / Page:	32 / 33
Firma / Organization:	T3 VAKFI
Tarih / Date	15.07.2025



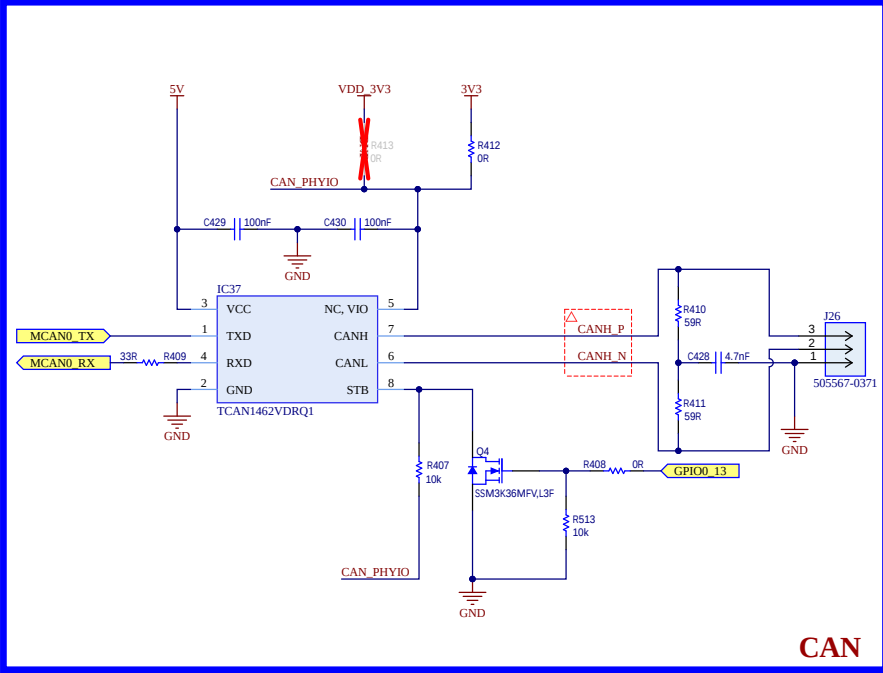
IMU 9-Axis



Humidity & Temperature



Pressure Sensor



CAN